

Improved cost models for optimizing CO₂ pipeline configuration for point-to-point pipelines and simple networks



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ABSTRACT

In this study, a new cost model is developed for CO₂ pipeline transport, which starts with the physical properties of CO₂ transport and includes different kinds of steel grades and up-to-date material and construction costs. This pipeline cost model is used for a new developed tool to determine the configuration leading to the lowest levelized costs for CO₂ transport, for point-to-point pipelines as well as for simple networks on different types of terrain and for different time frames. The model optimizes inlet pressure, diameter, steel grade and number of pumping stations.

Results show that gaseous CO₂ transport can give lower levelized costs than liquid CO₂ transport for point-to-point pipelines and for simple networks, if the CO₂ is stored in a reservoir with a low required injection pressure, like depleted natural gas fields. However, for storage fields with a required injection pressure of 8 MPa or higher (like aquifers), CO₂ can be better transported in a liquid form. For onshore pipelines transporting liquid CO₂, the optimal inlet pressure is 9–13 MPa and pumping stations are installed roughly every 50–100 km. For offshore pipelines, pumping stations are not an option and the inlet pressure is determined by the length of the pipeline. The maximum inlet pressure is about 25 MPa and for even longer pipelines, a larger diameter is selected. The levelized costs (excluding initial compression) for transporting 100 kg/s (about 3 Mt/y) over 100 km are between 1.8 and 3.3 €/t for liquid and 4.0–6.4 €/t for gaseous CO₂ transport. For larger mass flows the costs are decreasing, for instance transporting 200 kg/s (about 6 Mt/y) over 100 km are 1.2–1.8 €/t for liquid and 3.0–3.8 €/t for gaseous CO₂ transport.

Furthermore, results show that higher steel grades lead to lower investment costs for onshore pipelines transporting liquid CO₂. Using X120 in the long term reduces the pipeline costs up to 17%. For gaseous CO₂ transport, lower steel grades (like X42 and X52) are the best option. Also offshore pipelines do not benefit from the development of higher steel grades over time because the thickness should be at least 2.5% of the outer diameter.

The results indicate that oversizing the pipeline, to transport CO₂ from an additional source that is coming available later, is not always cost-attractive. This strongly depends on the time span of which further CO₂ sources are available and on the mass flows. Oversizing appears earlier cost-attractive compared to two point-to-point pipelines if the source with the largest mass flow becomes available first.

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1. Introduction

Carbon dioxide capture and storage (CCS) is a CO₂ abatement option that can contribute significantly to the reduction of CO₂ emissions to limit temperature increase (IEA, 2010; Moomaw et al., 2011; European Commission, 2011). With CCS, CO₂ from industrial and energy-related sources is captured and subsequently transported to an underground geological storage field. First estimations indicate that worldwide around 200,000–360,000 km pipeline is

required to store about 10 Gt/y in 2050 (IEA, 2009). This is a very significant amount, especially in comparison with the current CO₂ pipeline network of about 6000 km, which is mainly situated in the United States for enhanced oil recovery purposes (Mohitpour et al., 2012). Extending the CO₂ transportation network to the necessary scale would require a large planning effort and initial investments of about 400–750 billion euro have been reported (IEA, 2009). In the last decade, different economic models have been developed to assess the costs of CO₂ pipeline transport. A compile is presented in Knoope et al. (2013).

However, these cost models have two main shortcomings as indicated by a previous study (Knoope et al., 2013). Firstly, almost all models use current natural gas pipelines as the basis for their

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Nomenclature

CA	corrosion allowance
CCS	CO ₂ capture and storage
$C_{material}$	material costs for the pipeline
CRF	capital recovery factor
C_{steel}	steel cost
E	longitudinal joint factor
E_{pump}	energy consumption of pumping
EC_{pump}	energy costs of pumping stations
EC_{comp}	energy costs of compressor
E_{comp}	energy consumption of compression
F	design factor of the pipeline
f	Darcy–Weisbach friction factor
FEED	front end engineering design
FERC	Federal Energy Regulatory Commission
g	gravity constant
H	annual operation hours
i	in a certain year
I_0	base costs
I_{comp}	investment costs of compressor
I_{pipe}	investment costs pipeline
I_{pump}	investment costs of pumping station
ID	inner diameter
$ID_{calculated}$	calculated inner diameter
ID_{NPS}	inner diameter of the nominal pipe size
L	length of the pipeline
L_{pump}	maximum distance between pumping stations
LC	levelized cost of CO ₂ transport
LC_{ALL}	levelized costs including compression
LC_{trans}	levelized costs of transport excluding compression
m	CO ₂ mass flow
M	molecular mass of CO ₂
MAOP	maximum allowable operation pressure
me	multiplication exponent
n	number of units in parallel
N_{pumps}	number of pumping stations
NPS	nominal pipe sizes
O&M	operation and maintenance costs
OD	outer diameter
OD_{NPS}	outer diameter of the nominal pipe size
Offsh	offshore pipeline
OM_{pump}	O&M costs of pumping stations
OM_{com}	O&M costs of compressor
OM_{pipe}	O&M costs of pipeline
P_1	inlet pressure of the pump
$P_{1,x}$	inlet pressure of compression stage x
P_2	outlet pressure of the pump
$P_{2,x}$	outlet pressure of compression stage x
P_{ave}	average pressure in the pipeline
P_{inlet}	pressure inlet of the pipeline
Popu	populated terrain
P_{outlet}	pressure outlet of the pipeline
$P_{outlet.adapted.gas}$	adapted pressure outlet of the compressor for gaseous cases
$P_{outlet.last.pump}$	pressure outlet of the last pumping station
P_{max}	maximum pressure
PtP	point-to-point pipeline
R	universal gas constant
r	discount rate
Re	Reynolds number
ROW	right-of-way
S	minimum yield stress

S.Pop	sparsely populated terrain
T	inlet temperature of the compressor
t	thickness
T_{ave}	average fluid temperature
Tr	trunkline
UCCI	upstream capital cost index
US	United States
v	actual velocity
v_{max}	maximum velocity
v_{min}	minimum velocity
W_{pump}	capacity of pumping station
W_{comp}	capacity of the compressor
$W_{comp,0}$	base scale of the compressor
x	compression stage number
X_{stage}	total number of compression stages
y	scaling factor
z	lifetime
Z_{ave}	average fluid compressibility
Z_x	compressibility factor of CO ₂ in stage x
ΔP_{act}	actual pressure drop
ΔP_{design}	design pressure drop
Δz	height difference between in- and outlet of the pipeline
κ_x	specific heat ratio of the CO ₂ in stage x
η_{iso}	isentropic efficiency of the compressor
η_{mech}	mechanical efficiency of the compressor
η_{pump}	efficiency of the pumping station
μ	viscosity
ρ	density of the flow
ρ_{steel}	density of steel
ε	roughness height

cost estimation, thereby ignoring the different characteristics of CO₂ pipelines. The most obvious difference between natural gas and CO₂ pipelines is the higher operation pressure required by the latter. A higher operation pressure means that the wall thickness should be increased or that higher strength steel should be used, which will increase costs. So far, this effect is only taken into account by cost models based on the weight of the pipeline (Piessens et al., 2008; Gao et al., 2011).

Secondly, none of the costs models take into account the development in steel grades (Knoope et al., 2013), although some of the cost models are used for planning of a CO₂ infrastructure over time (Piessens et al., 2008; Van den Broek et al., 2010; ElementEnergy, 2010). Over the years, higher steel grades are and will be developed which achieve the same strength with a lower wall thickness (Felber and Loibnegger, 2009). However, it has still to be proven if very high steel grades have the required toughness to avoid crack propagation (Brauer et al., 2004). If they would have the required toughness, lower costs could be realized despite the higher costs per metric ton of steel, because less steel is required, transportation costs are lower and thinner walls are easier to weld (Gräfe et al., 2003; Cayrade, 2004; Felber and Loibnegger, 2009). The first aim of this study is to develop a cost model for CO₂ pipeline transport, related to the physical properties of CO₂, which include different types of steel grades and use up-to-date material and construction costs.

An additional issue identified by Knoope et al. (2013), was the large uncertainty in the costs and placement of pumping stations.¹ The installation (and location) of pumping stations is an

¹ In several publications, pumping stations are called booster stations.

economic decision resulting from tradeoffs between enlarging the diameter of the pipeline, increasing the inlet pressure or placing a pumping station. To determine whether installation of one or multiple pumping stations makes economically sense, Zhang et al. (2012) developed a techno-economic tool to analyze the number of pumping stations and the diameter for point-to-point CO₂ pipelines in China. However, they assume a fixed inlet pressure of 13.8 MPa and adapt it to a lower level if no pumping stations are present. Another limitation of the model of Zhang et al. (2012) is that different kind of terrains like offshore, sparsely populated or populated areas are not taken into account. This would change the outline of the system, because safety requirements are higher for pipelines in populated areas and pumping stations are very expensive offshore since a platform with energy supply is required (IEA GHG, 2002). Offshore CO₂ transport is currently considered very relevant in Europe, because offshore CO₂ storage is socially more accepted than onshore and two-thirds of the potential storage locations are situated offshore (EU GeoCapacity, 2008).

In the future, it is expected that trunklines will be installed transporting CO₂ from multiple sources to one or more sinks, thereby generating economies of scale (Chandel et al., 2010). A recent study indicates that almost 60% of potential CCS locations worldwide could benefit from trunkline configurations, leading to an estimated reduction of about 25% in total pipeline length (ElementEnergy, 2010). Furthermore, trunklines would smooth the CO₂ flow variations caused by intermittency and can lead to more operational flexibility than stand-alone pipelines (Fimbres Weihs and Wiley, 2012).

ElementEnergy (2010) analyses under which conditions a trunkline is more cost-effective than point-to-point pipelines. They conclude, for instance, that if two sources with an equal mass flow are 100 km away from the sink, it is more cost-effective to have point-to-point pipelines if the angle between source I–sink–source II is more than 60°. Other models analyze not only in which circumstances a trunkline is cost-effective but also how the overall network will develop and where trunklines would arise (among others: Middleton and Bielicki, 2009; Van den Broek et al., 2010; Morbee et al., 2012; Fimbres Weihs and Wiley, 2012). Middleton and Bielicki (2009), for instance, developed a spatial tool which determines not only where to build pipelines, but also selects the sources and sinks where to capture and store the CO₂. In an updated version, the development of the network is analyzed over time (Middleton et al., 2012). Also Van den Broek et al. (2010) include time and spatial items. They linked the tool to an energy-bottom up model to see the influence of several policy related decisions on the implementation of CCS and the outline of the CO₂ infrastructure.

Most models do not take into account pumping stations because the average distance between sinks and sources is small (Dahowski et al., 2004) or to simplify the modeling process (Middleton and Bielicki, 2009; Morbee et al., 2012). Others include pumping stations, but they use a simplified approach by fixing the distance between pumping stations (Wildenborg et al., 2004; Van den Broek et al., 2010). Two models incorporate pumping stations if the maximum or minimum pressure would be crossed but they use a fixed velocity (ElementEnergy, 2010) or use a maximum specific pressure drop to determine the diameter (Fimbres Weihs and Wiley, 2012). These kinds of assumptions lead to a specific diameter and it is not certain that this is the most cost-effective one.

An additional point to note is that all mentioned models assume that the CO₂ will be transported in supercritical or dense phase throughout the whole network.² For low mass flow rates and short

distances, however, it may be more cost-effective to transport the CO₂ (partly or contemporary) in the gaseous phase. Gaseous CO₂ transport is considered in a study of the IEA GHG (2007), in a study to CO₂ transport and storage in the Netherlands (EBN and Gasunie, 2010), in the CCS project in the Yorkshire and Humber area (Yorkshire Forward, 2008) and was considered for the onshore part of the Longannet project (ScottishPower CCS Consortium, 2011) and the initial phase of the Kingnorth project (E.ON, 2011a).³

To summarize, none of the existing models in literature performs a comprehensive economic minimization for CO₂ pipeline transport with respect to diameter, steel grade, wall thickness, number of pumping station, and inlet pressure incorporating both gaseous and liquid CO₂ transport. Therefore, the second aim of this article is to develop a model including these parameters, which would lead to the most cost-effective solution and configuration for CO₂ pipeline transport. This is done for point-to-point pipelines as well as for simple networks for different types of locations (offshore, sparsely populated, or populated) and for different time frames, namely the short (2020), medium (2030) and long term (2040).

2. CO₂ properties for pipeline transport

The critical point of pure CO₂ is at 31.1 °C and 7.4 MPa, see Fig. 1. CO₂ pipeline transport is often proposed in the dense phase, with pressure above 7.4 MPa but temperatures below 31 °C. Insulation or heating of the pipeline to ensure that the CO₂ is in the supercritical phase does not have an added benefit for the transportation chain and is not cost-effective according to Zhang et al. (2006, 2012). However, it can be considered to meet specific storage requirements.⁴ Nonetheless, insulation or heating of the pipeline is not taken into account in this study.

The operation regions, which are used in this study, are depicted in Fig. 1. In this article, it is assumed that the CO₂ is dehydrated, (near) pure and at a constant temperature of 15 °C (288 K) for onshore pipelines and of 4 °C (277 K) for offshore pipelines. The corresponding density, viscosity, compressibility factor and specific heat ratio are based on (NIST, 2012). A mixture of gas and liquid CO₂, a so-called two-phase flow, should be avoided in normal operation to avoid boiling, cavitation and operational problems with compressors and pumps (Skovholt, 1993; Svensson et al., 2004; ElementEnergy, 2010; Knoope et al., 2013). Note that a two-phase flow can still exist for pipeline of short distances (for instance in the well), for a short amount of time (during start-up/shut-down) or in some cases to meet specific storage requirements (DNV, 2010). For normal operation, the minimum pressure level is set to 8.0 MPa for liquid CO₂ transport. For gaseous transport, pressures between 1.5 and 3 MPa are allowed. Pressures higher than 3 MPa are not considered to avoid a two-phase flow, while pressures lower than 1.5 MPa are not incorporated because pressure is needed to ensure that the CO₂ flows.

³ Gaseous CO₂ transport was proposed for the first stage of the Kingsnorth project due to the low initial reservoir pressure of less than 0.3 MPa (E.ON, 2011a). In the Longannet project, gaseous CO₂ transport was considered because part of the existing onshore pipeline would not be suitable for liquid CO₂ transport (namely the maximum operation pressure of the existing pipeline is 8.5, 8.4 and 7.0 MPa).

⁴ In the ROAD project, the Netherlands, the low initial reservoir pressure of <3 MPa, makes it necessary to transport the CO₂ as a gas or throttle the liquid CO₂ before injection. Gaseous transport was excluded because this would lead to a too high pressure drop in the well. To avoid too low temperatures after throttling the liquid, which can cause hydrates in the well, the CO₂ has to be heated at the platform or the pipeline has to be insulated. The last option is chosen due to the limited length of the pipeline (26 km) which makes it affordable (Read, 2012).

² Dense phase is not a well-defined term. In this article, dense CO₂ refers to CO₂ above the critical pressure (7.4 MPa) independent of temperature, while supercritical CO₂ refers to CO₂ above the critical pressure and critical temperature (31.1 °C).

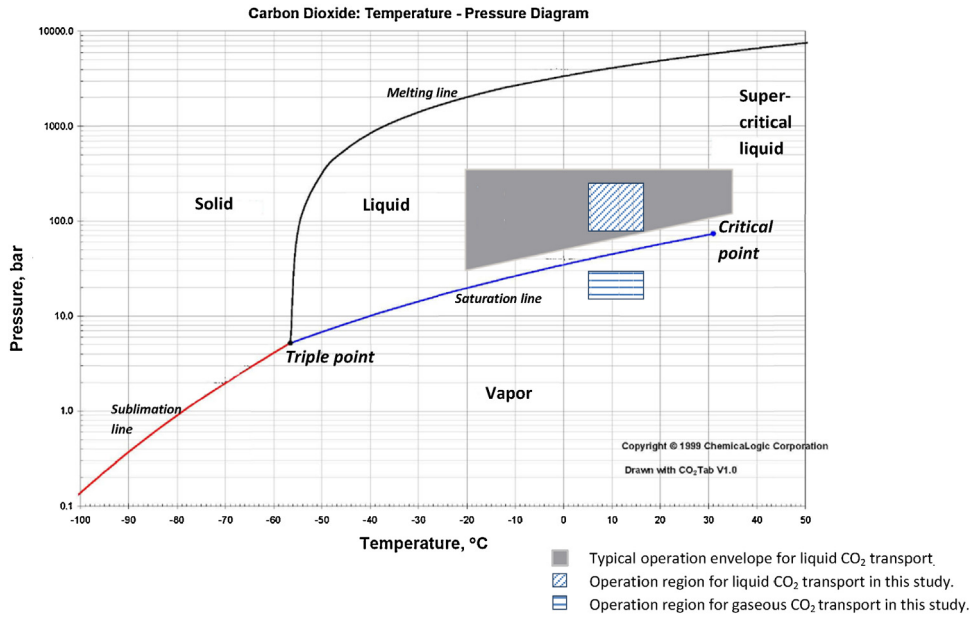


Fig. 1. Phase diagram for pure CO₂ (adapted from ChemicalLogic, 1999).

3. Description of the cost minimization process

In this study, a new cost model and an economic minimization tool are developed for CO₂ pipeline transport. All costs mentioned in this study are corrected with the upstream capital cost index (UCCI) to €₂₀₁₀ (IHS, 2013). Note that the UCCI is only valid for costs in dollars. Therefore, costs in Euros are first converted to dollars with the average exchange rate of the year where the costs are specific for, subsequently they are converted to \$₂₀₁₀ and then back to euros with the average exchange rate of 2010, which is 0.75€₂₀₁₀/\$₂₀₁₀ (OANDA, 2011).

In this study, a cost minimization process is executed, see Eq. (1) for the objective function. The decisions variables are the inlet pressure (P_{inlet}), a set of steel grades, number of pumping stations (integer variable, 0–10) and a set of outer diameters (OD_{NPS}). The main inputs for the cost minimization process are the mass flow (m), length of the pipeline (L), terrain type (offshore, populated or sparsely populated) and time frame (short, medium or long term). Additional explanation of the relations and the equations are given in the sections below:

is the CO₂ mass flow (kg/s); H are the number of operation hours (8760 h/y); P_{inlet} and P_{outlet} are the inlet and outlet pressure of the pipeline, respectively (Pa); P_{max} is the maximum pressure for CO₂ transport, which is 3.0 MPa for gaseous, 24 MPa for onshore liquid CO₂ and 35 MPa for offshore liquid CO₂ transport; N_{pumps} is the number of pumping stations; OD_{NPS} is the outer diameter related to the nominal pipe size; ΔP_{act} is the actual pressure drop (Pa/m); v , v_{min} and v_{max} is the actual, minimum and maximum velocity, respectively (m/s).

Although the objective function remains (almost⁵) the same, slightly different methodologies are required for different types of pipelines. A distinction can be made between pipelines over a single type of terrain and pipelines crossing multiple terrains, like sparsely populated and populated terrain or onshore and offshore. Furthermore, pipelines can be categorized into point-to-point pipelines which transport CO₂ from one source to one sink and a network which transport CO₂ from different sources to one or more sinks. With networks, the timing when capture units are expected to become available is very relevant as there maybe cases where it

$$LC_{low} = \text{Min} \left(\frac{\text{CRF} * (I_{pump} + I_{comp}) + \text{CRF} * I_{pipe} + OM_{pump} + OM_{pipe} + OM_{comp} + EC_{pump} + EC_{comp}}{m * H * 3.6} \right) \quad (1)$$

$$I_{pump} \rightarrow \varphi(P_{inlet}, P_{outlet}, N_{pumps}, m, L)$$

$$I_{comp} \rightarrow \varphi(P_{outlet}, m)$$

$$I_{pipe} \rightarrow \varphi(L, P_{inlet}, OD_{NPS}, \text{terrain type; time frame})$$

$$OD_{NPS} \rightarrow \varphi(m, v, \Delta P_{act}, \text{terrain type})$$

Subject to:

$$v_{min} < v < v_{max}$$

$$P_{outlet} < P_{inlet} < P_{max}$$

$$P_{outlet} = P_{inlet} - \Delta P_{act} * L / (N_{pumps} + 1)$$

where LC are the levelized cost of CO₂ transport (€/t CO₂); CRF is the capital recovery factor; $I_{pump/pipe/comp}$ and $OM_{pump/pipe/comp}$ are the investment and operation and maintenance (O&M) costs of pumps, pipeline and compressor, respectively (€); $EC_{pump/comp}$ are the energy costs of pumps and compressor, respectively (€/y); m

is more cost-attractive to construct two point-to-point pipelines rather than a trunkline. Fig. 2 depicts an outline of the methodology.

3.1. Cost minimization of one pipeline in a single type of terrain

Gaseous as well as liquid CO₂ transport are included in the model. For gaseous CO₂ transport, the inlet pressure ranges from 1.6 to 3 MPa, in steps of 0.1 MPa, while the outlet pressure is fixed on 1.5 MPa. This low outlet pressure makes it only interesting to store the CO₂ in a storage field with a low reservoir pressure, like a depleted natural gas field. The possibility of recompressing the CO₂ along the pipeline is not included for gaseous CO₂ transport

⁵ The formula for the levelized costs is different if timing aspects play a role, see Section 3.4.

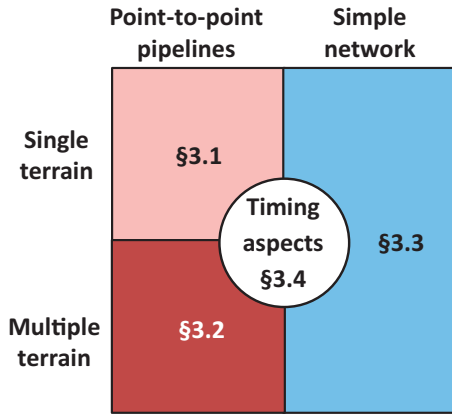


Fig. 2. Outline of the methodology for different types of pipelines.

due to the high energy consumption and recompression costs.⁶ For onshore liquid CO₂ transport, the inlet pressures range from 9 to 24 MPa, in steps of 1 MPa, and with 0–10 pumping stations. For offshore liquid CO₂ transport, pumping stations are not an option and inlet pressures range from 9 to 35 MPa, in steps of 1 MPa. All liquid cases have a fixed outlet pressure of 8.0 MPa. For all cases, the configuration and the levelized costs are calculated. In Fig. 3, the steps of the cost minimization process are presented in a flow diagram. Additional equations are given in the Appendix.

For the cases transporting liquid CO₂, the specific design pressure drop is calculated by taking into account the in- and outlet pressure, number of pumping stations and the height difference of the trajectory (Eq. (2)). This specific design pressure drop is used to calculate the required inner diameter ($ID_{calculated}$), see Eq. (3). Since the density varies considerably between the in- and outlet of gaseous CO₂ transport, Eq. (4) is used for calculating the inner diameter⁷ (Mohitpour et al., 2003; McCoy and Rubin, 2008; Chandel et al., 2010):

$$\Delta P_{design} = \frac{(P_{inlet} - P_{outlet}) * (N_{pumps} + 1)}{L} + \frac{g * \rho * \Delta z}{L} \quad (2)$$

$$ID_{calculated_liquid} = \left(\frac{8 * f * m^2}{\pi^2 * \rho * \Delta P_{design}} \right)^{\frac{1}{5}} \quad (3)$$

$$ID_{calculated_gaseous} = \left(\frac{-16 * Z_{ave}^2 * R^2 * T_{ave}^2 * m^2 * f * L}{\pi^2 [M * Z_{ave} * T_{ave} * R * (P_{outlet}^2 - P_{inlet}^2) + 2 * g * P_{ave}^2 * M^2 * \Delta z]} \right)^{\frac{1}{5}} \quad (4)$$

$$P_{ave} = \frac{2}{3} \left(P_{outlet} + P_{inlet} - \frac{P_{outlet} * P_{inlet}}{P_{outlet} + P_{inlet}} \right) \quad (5)$$

where ΔP_{design} is the design pressure drop (Pa/m); P_{inlet} is the pressure inlet (Pa); P_{outlet} is the pressure outlet (Pa); N_{pumps} is the

number of pumping stations; L is the length of the pipeline (m); g is the gravity constant (9.81 m/s²); ρ is the density (kg/m³)⁸; Δz is the height at the inlet of the pipeline minus the height of the outlet of the pipeline⁹ (m); $ID_{calculated_liquid}$ is the inner diameter of the pipeline transporting liquid CO₂ (m); f is the Darcy–Weisbach friction factor (see Appendix for equation); m is the CO₂ mass flow (kg/s); $ID_{calculated_gaseous}$ is the inner diameter of the pipeline transporting gaseous CO₂ (m); Z_{ave} is the average fluid compressibility¹⁰; R is the Universal gas constant (8.31 Pa m³/mol K); T_{ave} is the average fluid temperature (K); M is the molecular mass of CO₂ flow (kg/mol) and P_{ave} is the average pressure in the pipeline.

Pipelines are available in so called nominal pipe sizes (NPS).¹¹ NPS are related to the outer rather than the inside diameter, thus the next available outer diameter of the NPS (OD_{NPS}) is selected with respect to $ID_{calculated}$. If the calculated diameter is larger than the largest NPS included in the model, then the specific combination of inlet pressure and pumping stations is not taken into account. The possibility of placing multiple pipelines next to each other is currently not considered.

To relate OD_{NPS} to the inner diameter of NPS (ID_{NPS}), the thickness is calculated (see Eq. (6)) assuming that the maximum allowable operation pressure is 10% higher than the (steady state) inlet pressure and subsequently rounded up to 0.1 MPa.¹² To ensure that the wall thickness does not become too thin, a maximum OD_{NPS}/t ratio of 100 for onshore pipelines (Gräf et al., 2003) and 40 for offshore pipelines (Hillenbrand et al., 2001; DNV, 2012)¹³ have been set. The calculated wall thickness is rounded up to half or whole millimeters:

$$t = \frac{OD_{NPS} * MAOP}{2 * S * F * E} + CA \quad (6)$$

where t is the thickness (m); OD_{NPS} is the outer diameter of the nominal pipe size (m); $MAOP$ is the maximum allowable operation pressure (MPa); S is the minimum yield stress (MPa); F is the design factor related to the terrain and population density; E is the longitudinal joint factor (=1) and CA is the corrosion allowance (=0.001 m).¹⁴

⁸ The density is based on the lowest pressure in the pipeline (P_{outlet}) to take a conservative approach. The temperature of the CO₂ is 4 °C for offshore pipelines and 15 °C for onshore pipelines.

⁹ The formula gives only correct results if the height difference is more or less equally divided at the route. For instance, if the pipeline needs to go up a mountain, a pumping station may be needed to compensate the strong pressure drop due to gravity, while when going down a mountain, pressure reducing equipment may be needed. Hence, for such cases the route has to be split up into segments with comparable slopes.

¹⁰ The average fluid compressibility factor is based on the average pressure and the temperature of the fluid (which is in this study 15 °C for onshore pipelines and 4 °C for offshore pipelines).

¹¹ The following NPS related to outer diameter (OD_{NPS}) are included for onshore and offshore pipeline transport: 0.11, 0.17, 0.22, 0.27, 0.32, 0.41, 0.51, 0.61, 0.76, 0.91, 1.07 and 1.22 m. For onshore pipeline transport, OD_{NPS} of 1.32 and 1.42 m are also included.

¹² The $MAOP$ is often higher than the inlet pressure. A higher $MAOP$ leads to a higher wall thickness and higher capital costs, but it gives some operational freedom. Moreover, it is to ensure that the $MAOP$ is not crossed in all circumstances. For instance, during an emergency shutdown of a block valve at the end of the pipeline, shock waves are formed which increase the pressure significantly. Also operating the pumping stations with a lower mass flow can lead to a higher inlet pressure. These issues have to be taken into account if a (CO₂) pipeline is designed for construction. However, for the purpose of this study, a standard 10% rule-of-thumb is used.

¹³ The wall thickness in offshore pipeline is larger to prevent collapse of the pipeline due to the external pressure of the water. In the DNV standard for offshore pipelines a OD/t ratio between 15–45 is stated (DNV, 2012). In Hillenbrand et al. (2001), a relation between water depth and OD/t ratio is given, where a ratio of 40 relates to a water depth of 400 m.

¹⁴ For pure CO₂ transport without free water no, or very limited, corrosion is expected in the pipeline. Field experience of CO₂ pipelines installed for enhanced oil recovery in the US, show a low corrosion rate of 0.25–2.5 μm/y (Cole et al., 2011).

⁶ Initial runs showed that increasing the diameter results in lower levelized costs than adding a recompression station for pipelines transporting gaseous CO₂. For instance, for a gaseous pipeline of 100 km transporting 300 kg/s, the lowest levelized costs of 10.0 €/t are realized with an inlet pressure of 2.3 MPa and a diameter of 1.22 m. If one intermediate compressor is installed, the levelized costs increase to 10.2 €/t, while the optimal diameter remains 1.22 m and the inlet pressure decreases to 2.0 MPa. By running cases for various mass flows and lengths, it was concluded that installing an intermediate compressor increase the levelized costs and it is more cost-effective to increase the inlet pressure or even installing a larger diameter.

⁷ In principle, Eq. (4) can be used for liquid as well as for gaseous cases. However, for liquid cases Eq. (3) is used for simplicity.

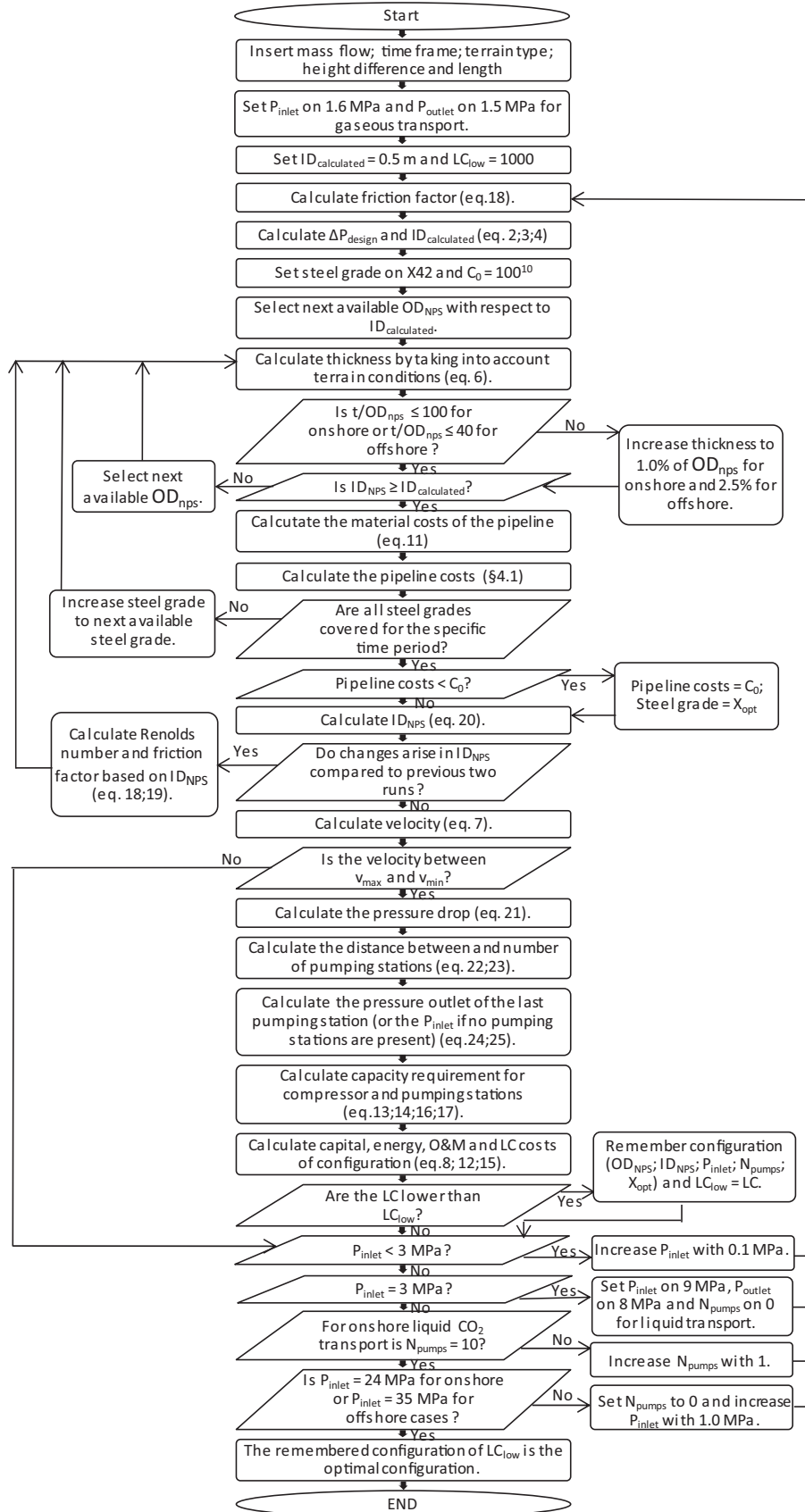


Fig. 3. Flow diagram for a point-to-point pipeline over one kind of terrain.

The thickness depends on a design factor related to the location of the pipeline and the yield stress of the material. A design factor of 0.72 is used for pipelines in areas having a low population density (sparsely populated areas) and for offshore pipelines (Code of Federal Regulation, 2010). For areas having a high population density, the wall thickness of a pipeline should be higher for safety reasons and, therefore, a lower design factor is adopted (Code of Federal Regulation, 2010).¹⁵ In this study, a design factor of 0.5 is used for onshore areas with a higher population density (populated areas).

To ensure that the configuration is feasible, the velocity is calculated with the selected ID_{NPS} (Eq. (7)). The velocity has to be below a certain value to avoid erosion, vibrations and damaging of the pipeline. In the NORSOK standard, a maximum velocity of 6 m/s is specified in carbon steel pipelines for the transport of a liquid without solid particles, which would be the case with CO₂ transport (NORSOK, 2006). A minimum velocity is not specified in the NORSOK standard. In this article, a velocity range of 0.5–6 m/s is assumed for liquid cases and of 5–20 m/s for gaseous transport. If a specific combination of inlet pressure, number of pumping stations and diameter gives a velocity outside the identified range, the case is ignored:

$$v = \frac{4 * m}{ID_{NPS}^2 * \pi * \rho} \quad (7)$$

where v is the actual velocity (m/s); m is the CO₂ mass flow (kg/s); ID_{NPS} is the inner diameter of the NPS pipeline (m); and ρ is the density (kg/m³).

The levelized costs of CO₂ transport for all cases are calculated with Eq. (8) and with the economic assumptions given in Table 1. The case with the lowest levelized costs is the most cost-effective combination of inlet pressure, diameter, steel grade and number of pumping stations:

$$LC = \frac{CRF * (I_{pump} + I_{comp}) + CRF * I_{pipe} + OM_{pump} + OM_{pipe} + OM_{comp} + EC_{pump} + EC_{comp}}{m * H * 3.6} \quad (8)$$

$$CRF = \frac{r}{1 - (1 + r)^{-z}} \quad (9)$$

where LC are the levelized cost of CO₂ transport (€/t CO₂); CRF is the capital recovery factor; $I_{pump/pipe/comp}$ and $OM_{pump/pipe/comp}$ are the investment and operation and maintenance (O&M) costs of pumps, pipeline and compressor, respectively (€); $EC_{pump/comp}$ are the energy costs of pumps and compressor, respectively (€/y); m is the CO₂ mass flow (kg/s); H are the number of operation hours (8760 h/y); r is the discount rate (%); and z is the lifetime (years).

¹⁵ In the US Code of Federal Regulation, four different types of areas each with a different design factor are distinguished (Code of Federal Regulation, 2010). First, an area which has less than 10 buildings for human occupancy placed within 200 m of each side of the pipeline for any continuous mile (1.6 km) of pipeline length. This type of area would be designed with a design factor 0.72. Second, a design factor of 0.6 is used for an area with more than 10, but fewer than 46 buildings, intended for human occupancy in an area of 200 m around the pipeline for any continuous mile of pipeline length. Third, a design factor of 0.5 is used for an area which has more than 46 building for human occupancy in an area of 200 m on each side of the pipeline for any continuous mile (1.6 km). Fourth, a design factor of 0.4 is used in an area where there are buildings with four or more stories above the ground present in the near surrounding of the pipeline (<200 m). In this study, the first type, referred to with sparsely populated terrain, and the third type, referred to with populated area, are incorporated.

Table 1
Economic assumptions.

Parameter	Unit	Base value	Sensitivity
Design lifetime of the pipeline	Years	50	
Design lifetime of compressors and pumping stations	Years	25	
Interest rate	%	10	5–15
Operation hours	h/y	8760 ^a	
O&M costs compressors and pumping stations	%	4.0	
O&M costs pipeline	%	1.5	
Electricity costs ^b	€/MWh	100	50–150
Steel factor ^{c,d}		1	0.5–1.5
Labor cost ^{d,e}	€/m ²	825	413–1650

^a In practice, the operation time would be <8760 h/y. However, if multiple sources are combined in a pipeline, the pumping stations will operate constantly since it is unlikely that all plants will shut down at the same time.

^b The electricity cost is not constant over time. However, for simplicity it is assumed that the costs will remain similar. An electricity cost of about 100€/MWh is projected for power plants with CCS (ZEP, 2011). However, there is significant uncertainty in the costs of CCS power plants and hence in the electricity cost. In the ZEP report (ZEP, 2011), a literature overview is made of the electricity costs with CO₂ capture and electricity costs range from 42 to 157€/MWh for a Nth of a kind plant. In this study, a sensitivity analysis is conducted for electricity costs of 50 and 150€/MWh.

^c The steel costs used in this study are from April 2012. In the first eight months of 2012 the average CRU global steel price index (CRUspi) was 191 (CRU, 2012). In the period 2000–2012, the CRUspi varied between 69 and 293 (CRU, 2012). For the sensitivity analysis, a multiplication factor of 0.5–1.5 is used. This leads, for instance, to a steel cost range of 0.53–2.27€/kg for X80.

In principle, the steel costs would increase if strict climate change policies come in place. It is estimated that the CO₂ capture costs for the steel industry are 45–73€/2010/t CO₂, the avoided CO₂ emissions are 0.3–0.8 t CO₂/t steel and the realized CO₂ emissions are 0.9–1.5 t CO₂/t steel in the short and medium term (Kuramochi et al., 2012). This would increase the steel costs with 0.01–0.06€/kg due to the CCS equipment and with an additional 0.09–0.15€/kg if a tax of 100€/t emitted CO₂ is raised. Hence, the steel costs would increase with about 5–15% if strict climate policies and CCS are applied in the steel industry. This increase is considerably less than the historical variation and is, therefore, not taken into account further.

^d The miscellaneous costs are a percentage of the labor and material costs, and therefore they will also increase if the labor or steel costs are increased.

^e The range of labor costs is calculated from the FERC cost database taking into account pipelines constructed in the period 2008–2012. Pipelines lengths lower than 10 miles are ignored, because they can have very high (labor) costs. The base value is the average, while the sensitivity range is 50–200% due to the higher uncertainty, see Section 4.1.2.

3.2. Cost minimization of a pipeline crossing different types of terrain

3.2.1. Pipeline through two terrain types

If the pipeline is installed partly in populated and partly in sparsely populated terrain, the cost minimization process described in Section 3.1 is first conducted by assuming that the pipeline was installed completely on sparsely populated terrain. Subsequently, it is assumed that the optimal configuration with respect to NPS and inlet pressure would not change if the pipeline is partly placed on populated terrain. As the NPS is related to outer diameter, the inner diameter of the pipeline passing through populated terrain will decrease because a larger thickness is required. Since the inner diameter is smaller, the actual specific pressure drop is higher. The higher pressure drop is assumed to be compensated by the last pumping station along the route, which is placed further away from the sink and will pump the CO₂ to a higher pressure level.

3.2.2. Pipeline transporting CO₂ from onshore to an offshore storage location

If the pipeline comprised of an onshore and offshore section, the route is divided in two parts, which are optimized separately and then combined. The outlet of the first, onshore, section is initially set to 8 MPa for liquid CO₂ transport. Subsequently, the cost

minimization process is followed as described in Section 3.1. After determining the most cost-effective case for the first section, the outlet pressure of the first section is used as inlet pressure for the second, offshore, section. For the second pipeline section, a cost minimization is conducted with inlet pressures ranging from 9 to 24 MPa and a fixed outlet pressure of 8 MPa. A pumping station between the onshore and offshore section can be installed to pump the CO₂ to a higher inlet pressure. In this study, it is assumed that no expansion devices are installed to reduce pressure. Hence, if the outlet pressure of the onshore section is higher than the inlet pressure of the offshore section, the case is eliminated.

The minimum levelized costs for the onshore section, plus the minimum levelized costs for the offshore section are the lowest possible cost solutions for the entire pipeline for an outlet pressure of 8 MPa of the onshore pipeline section. Subsequently, the outlet pressure of the onshore section is raised with 1 MPa and the same cost minimization process is performed again. This process is repeated until the outlet pressure of the onshore section becomes 23 MPa. The lowest possible levelized cost solutions of all outlet pressures are compared, and the lowest one is selected.

Since the inlet pressures of the offshore as well as of the onshore section are in whole MPa, this method is only suitable for long onshore and offshore pipelines of more than 100 km. Otherwise the influence of rounding becomes too large. For short distances onshore, the pipeline can be better optimized with the feeder approach described in Section 3.3.

For gaseous CO₂ transport, the same cost minimization process is followed as for liquid CO₂ transport. However, here the initial outlet pressure of the onshore section is set on 1.6 MPa, while the inlet pressure range from 1.7 to 3.0 MPa. The possibility to recompress the flow along the route is excluded. After each round, the outlet pressure of the onshore section is raised with 0.1 until 2.9 MPa. The case resulting in the lowest levelized costs for gaseous transport is compared with the one calculated for liquid transport and the lowest cost solution is selected.

3.3. Feeders to and distribution pipelines from the trunkline

If multiple pipelines (feeders) are combined to one large pipeline (a trunkline) or multiple distribution pipelines are used, then the cost minimization process gets too many variables. To limit the number of possibilities, the specific design pressure drop is assumed to be 10 Pa/m for gaseous and 30 Pa/m for liquid transport in the feeders and distribution pipelines. A sensitivity analysis is included to assess the consequences of a 50% lower or higher specific pressure drop. Furthermore, all feeders and distribution pipelines are assumed to be constructed from X80 in the short term, X100 in the medium term and X120 in the long term for liquid CO₂ transport, and X42 is assumed for gaseous transport for all time frames. The steel grade of the trunkline remains a result from the cost minimization process. Additionally, the possibility of installing pumping stations on feeders and distribution pipelines is not considered. All these simplifications would only have a minor influence on the total levelized costs because the feeders and distribution pipelines have short distances compared to the trunkline. For the configurations calculated in this study, the distribution lines are 10 km, the feeders 10–75 km and trunklines 100–1500 km.

3.3.1. Network options

The levelized costs of the following four network options are analyzed and the one resulting in the lowest levelized costs is selected. Additionally, breakeven distances for feeders are calculated between different network options to assess when certain network options are cost-effective.

- I. Gaseous transport in the feeders as well as in the trunkline and distribution pipelines.
- II. Gaseous transport in the feeders and liquid transport in the trunkline and distribution pipelines.
- III. Liquid transport in the entire network, where the streams flow immediately from the feeders into the trunkline without a pumping station.
- IV. Liquid transport in the entire network, having a pumping station installed just at the inlet of the trunkline.

For network option II, it is assumed that the feeders can have a lower MAOP than the trunkline because a compressor is placed in between them. However, for network option I, III and IV, the MAOP of the feeders, distribution pipelines and trunkline are assumed to be similar. In network option I and III, the feeders have a higher inlet pressure than the trunkline, and therefore the MAOP of the trunkline increases. It is assumed that by changing the MAOP and the thickness of the trunkline, the optimal diameter and steel grade would not change. However, the location and number of pumping stations may change due to the higher specific pressure drop and the higher operation pressure. Consequently, these and the costs of the configuration are calculated again.

For all network options, the trunkline is optimized by assuming a large compressor station at the beginning of the pipeline.¹⁶ For calculating the costs of the network configuration, the costs of the inlet compressor and pump are extracted, while the costs are added of the compressors installed at the capture sites and at the beginning of the trunkline, if applicable. The levelized costs of compression increase with about 1–4% because one large compressor is replaced by two smaller compressor stations. Nevertheless, the outcome of the cost minimization process is not expected to be significantly influenced by this simplification because for mass flows above 100 kg/s already two compressors have to be installed in parallel due to the maximum capacity of one compressor of 35 MW.

3.4. Timing

The timing when CO₂ from different sources is expected to become available is very relevant for building a network. It is possible to construct a trunkline which is optimized for the combined mass flows of the sources and, in this way, try to profit from economies of scale. Another option is to construct point-to-point pipelines from each source to the sink, which are optimized for their own mass flow and length. In this study, it is investigated which option is more cost-attractive for various distances between the sources, distances to the storage location, mass flows and time difference when the sources become available.

The outline of the trunkline and the point-to-point pipelines are calculated with the method presented in Section 3.1. The trunkline is over-dimensioned and consequently the specific pressure drop decreases. Assuming that the outlet pressure remains at the same level, less pumping stations have to be installed initially along the route. If no pumping stations are present along the pipeline, the inlet pressure can be decreased. When the trunkline is constructed, only pumping stations are installed which are required for the mass flow of source A. Furthermore, pumping stations are only installed at locations where later on, if the mass flow of source B is also transported by the trunkline, also pumping stations would be installed.

When the second source becomes available, a short feeder is built to connect the source to the trunkline and if additional pumping station capacity is required on the pipeline, this is also

¹⁶ The compressor is often included in the capture plant.

installed. In principle, the location of the trunkline could be anywhere. However, for simplicity it is assumed that the trunkline starts from the source which is first coming available, and the feeders are installed to the beginning of the trunkline.

For calculating the levelized costs of the point-to-point pipelines as well as for the trunkline approach, Eq. (8) cannot be used because the costs are not constant over the lifetime. To take this into account Eq. (10) is used. It is assumed that both sources will transport CO₂ 30 years. Hence, the trunkline transport the CO₂ from only the second source after 30 years. To make a fair comparison, also the point-to-point pipeline are assumed to have lifetimes of 30 years:

$$LC = \frac{\sum_i^z (I_{pump,i} + I_{pipe,i} + I_{comp,i} + OM_{pump,i} + OM_{pipe,i} + OM_{comp,i} + EC_{pump,i} + EC_{comp,i}) / (1 + r)^i}{\sum_i^z (m_i * H * 3.6) / (1 + r)^i} \quad (10)$$

where LC are levelized cost of CO₂ transport (€/t CO₂); z is the lifetime of the system (y); $I_{pump/pipe/comp,i}$ and $OM_{pump/pipe/comp,i}$ are the O&M and investment costs of pumps, pipeline and inlet compressor in year i , respectively (€); $EC_{pump/comp,i}$ are the energy costs of pumps and initial compression in year i , respectively (€/y); r is the discount rate (%); m_i is the mass flow in year i (kg/s) and H are the number of operation hours (8760 h/y).

4. Development of cost models for CO₂ transport

In this section a description is provided of the cost models which were developed for the three main elements of a pipeline system, namely the inlet compressor, pipeline, and pumping stations. A literature overview of the cost models for CO₂ pipeline transport and pumping stations is given in Knoope et al. (2013). Furthermore, key cost model characteristics are identified for pumping stations and pipeline for a system analysis over time in Knoope et al. (2013). These are used in the development of the cost models below.

4.1. Pipeline

4.1.1. Material costs

Material costs should be based on the weight of the pipeline, see Eq. (11) (Knoope et al., 2013). The material costs are related to the steel grade of the pipeline, where steel grades with a higher yield stress are more expensive per kilogram, see Table 2. Steel grades up to X70¹⁷ are widely used, while X80 has currently more than 1000 km of experience (Hillenbrand et al., 2008), including at least one CO₂ pipeline (Pipeline and Gas Journal, 2009). Although X80 is not yet the dominate pipeline material today, it is expected that the number of X80 pipelines will increase in the next years to more than 15,000 km in total (Spinelli et al., 2011). In this article, it is assumed that X80 is available in the short term, while steel grades up to X100 will be available in the medium term and X120 is expected to become available in the long term:

$$C_{material} = t\pi * (OD_{NPS} - t) * L * \rho_{steel} * C_{steel} \quad (11)$$

where $C_{material}$ is the material costs for the pipeline (€); t is the thickness (m); OD_{NPS} is the outer diameter of the nominal pipe size (m); L is the length of the pipeline (m); ρ_{steel} is the density of steel, which is 7900 kg/m³ for all steel grades; and C_{steel} are the steel cost (€/kg).

4.1.2. Labor costs

Average labor costs for onshore pipelines are 825€/m² (21€/inch/m) with a standard deviation of 420€/m²

(11€/inch/m),¹⁸ based on FERC data¹⁹ from the period 2008–2012. These labor costs are valid for developed countries. In developing countries, the labor costs are lower and the labor costs could be corrected with a location factor of, for instance, the IEA GHG (2002).

For offshore pipelines, the labor costs are generally higher than for onshore pipelines because special pipelay barges and risers are needed, the onshore–offshore landfall is costly and the connection to an injection point is more complicated than onshore. These costs are estimated at 35 M€ and independent of pipeline length²⁰ (Austell et al., 2011). For offshore pipelines, less cost data is

available than for onshore pipelines and only one offshore pipeline is included in the FERC database, that is constructed after 2008 and longer than 10 miles. After reducing the labor costs with the fixed amount of 35 M€, this pipeline has estimated variable labor costs of 845€/m². These costs are comparable with the estimated variable labor costs of 830€/m² of the Nord Stream project, which are two very long parallel natural gas pipelines in the Baltic Sea²¹ (Nord Stream, 2012). These estimated variable labor costs are almost similar to the average labor costs extracted from the FERC data for onshore pipelines, and therefore, labor costs of 825€/m² are used for onshore as well as offshore pipelines. However, it has to be kept in mind that the offshore labor costs are only based on two data points and, hence, the labor costs for offshore pipelines are even more uncertain than the ones for onshore pipelines.

4.1.3. Right-of-way and miscellaneous costs

Almost the same amount of land is needed for construction of a small or a large diameter pipeline. Therefore, a fixed amount per meter length is used for the right-of-way (ROW) costs, like in Bureau et al. (2011). From the FERC database of the last 5 years, an average amount of 83€/m is extracted for onshore pipelines.²² Furthermore, 8 of the 14 offshore pipelines have zero ROW, as no land has to be acquisitioned. In this study the ROW costs for offshore pipelines are assumed to be zero (Bureau et al., 2011).

In this study, the miscellaneous costs are expressed as a fixed percentage of the material and labor costs, as was done in Bureau et al. (2011). From the FERC data of the last five years an average percentage of 25% is calculated, with a standard deviation of 10%. This percentage is used for onshore as well as offshore pipelines.

¹⁸ In the period 2010–2012, the average labor costs were 950€/m² (24€/inch/m) with a standard deviation of 350€/m² (9€/inch/m) and in the period 2002–2012, the average was 790€/m² (20€/inch/m) with a standard deviation of 490€/m² (12€/inch/m).

¹⁹ The FERC data is reported in the Oil and Gas Journal (True, 2002, 2003; True and Stell, 2004; Smith et al., 2005; Smith, 2006, 2007, 2008, 2009, 2010, 2011, 2012).

²⁰ In a network approach, where all pipelines are constructed simultaneously, the fixed costs are only incorporated once because the pipelay barge has to be (de)mobilized once. With building up a simple network over time, where the second pipeline, for instance, is constructed 5–10 years later than the first pipeline, the fixed costs are included twice.

²¹ The Nord Stream pipeline consist of two pipelines with a diameter of 1.22 m, each having a length of 1224 km. The total investment is 6.4 billion €₂₀₁₀ and 2.6 billion € is for the pipes and pipeline material (Nord Stream, 2012). The rest (3.8 billion €) consist of labor, miscellaneous and ROW costs. Since ROW costs are assumed to be zero for offshore pipelines and miscellaneous costs are assumed to be 25% of labor and material costs (see Section 4.1.3), it can be calculated that the labor costs are 2.5 billion €. Subtracting the fixed costs of 35 M€, result in variable costs of 830€₂₀₁₀/m².

²² The uncertainty in this amount is relatively large with a standard deviation of 70€/m. Nevertheless, the data is used because there is no better data available and the share of the ROW costs in the overall pipeline costs are only minor (with a maximum of about 15% for small sized pipelines).

¹⁷ The numbers behind the X refers to the yield strength of the material in ksi. Hence, X100 refers that the pipeline can have stress up to 100 ksi (=690 MPa) before the pipeline is non-reversible deformed.

Table 2Yield stress, costs and availability of different steel grades^a (AG Der Dillinger Hüttenwerke and GTS industries SA., 2012).

Steel quality (US standard)	Steel quality (EU standard)	Yield stress (MPa)	Steel costs (€ ₂₀₁₀ /kg)	Availability
X42	S275M ^b	275	1.17	Short term
X52	S355M ^b	355	1.20	Short term
X65	S460M ^b	460	1.37	Short term
X70	S500QL ^c	500	1.49	Short term
X80	S550QL ^c	550	1.51	Short term
X90	S620QL ^c	620	1.53	Medium term
X100	S690QL ^c	690	1.54	Medium term
X120	S890QL ^c	890	1.79	Long term

^a The costs are for heavy steel plates used in the steel construction, which are made of fine grained structural steel. It is assumed that these costs are representative for steel pipelines. Additional charges related to specific dimensions, testing and certification costs are ignored in this study. Costs for forming and welding the plate into a pipeline are not known and are therefore not taken into account. To analyze the costs of steel plates are comparable to the costs of steel pipelines, they are compared with the average costs of constructed double submerged arc-welded pipeline of ≥ 24 inch (≈ 0.61 m) indicated in Preston (2013). In 2012, the average pipeline costs were 1.26€₂₀₁₀/kg for a mix of applied steel grade (Preston, 2013). These costs are comparable to the costs of the lower steel grades given. To analyze the implications of a different steel price, the steel prices are included in a sensitivity analysis (see Section 5.7).

^b These steel qualities are thermo-mechanically rolled, leading to a steel with moderate yield stress, high toughness values and good weldability.

^c These steel qualities are quenched with water and tempered, resulting in high yield stresses. However, the toughness of the steel is lower than of thermo-mechanically rolled steel, increasing the chance of fracture propagation and can make crack arrestors necessary. Additionally, the costs of water quenching and tempering are higher than for thermo-mechanically rolled steel.

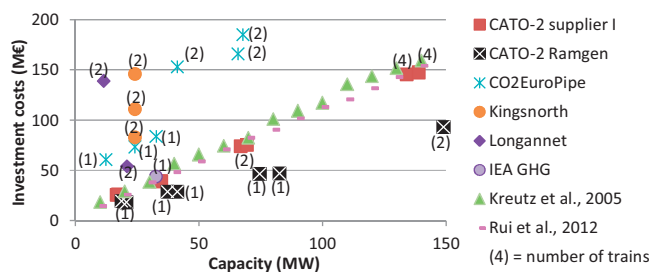


Fig. 4. Investment costs and capital costs for a compressor. The capital costs are based on (Kreutz et al., 2005; IEA GHG, 2008; Van Osch et al., 2010; ScottishPower CCS Consortium, 2011; Apeland et al., 2011a; E.ON, 2011a; Apeland et al., 2011b; Rui et al., 2012), see for an overview supplementary material. Note that the RAMGEN compressor is based on a new technology, which is not yet commercially available.)

4.2. Compressor

4.2.1. Capital costs

In Fig. 4, the investment costs are presented for CO₂ compressors of several front end engineering design (FEED) studies and vendor quotations. The specific investment costs vary significantly between the different estimations, but this is largely caused by the difference in assumed installation factors. These vary, for instance, from 1.5 for the cost estimation of the IEA GHG to 6.4 for several cost estimations of CO₂ Europipe.

Based on the available data it is difficult to extract a good correlation between capacity and costs due to the costs discrepancy and the limited amount of data points. Therefore, the cost estimation of Kreutz et al. (2005) is used in this study, which is close to the material costs provided by a supplier multiplied with a standard installation factor of 2.82.²³ The cost relation, which is a standard scaling formula, is given in Eq. (12). The costs of Kreutz et al., are quite old, but similar costs are found in a more recent study based on the costs of natural gas compressors in the US (Rui et al., 2012). The cost equation of the study of Rui et al. (2012) is not used, since the cost model is based on a quadratic equation which is simply the result of fitting and the parameters do not have any physical or economic meaning.

Kreutz et al. (2005, 2008) do not mention a maximum scale. However, from the installed capacities and number of trains in

Fig. 4, it can be estimated that the maximum scale of one compressor is about 35 MW_e.²⁴ Installing two or more compressors in parallel give a cost advantage via a multiplication factor, see Eq. (12) (Meerman et al., 2012).

Furthermore, in the source of Kreutz et al. (2005, 2008) it is rather unclear if the costs relation they develop include aftercooling. However, the flow diagrams indicate that the CO₂ after the compressor goes immediately to the pipeline and no additional cost for after cooling are included. Hence, it seems to be that the costs of aftercooling are already included in the compressor costs. The cost estimations provided by a supplier, which are almost similar to the costs relation of Kreutz et al., included aftercooling to about 30 °C. The rest of the cooling (to 10–15 °C) is assumed to happen in the pipeline. An inlet temperature of about 30 °C is not expected to be problematic for CO₂ transport as the studies of ROAD, Jän-schwalde and Kingsnorth all indicate higher inlet temperatures exceeding 40 °C and rather high maximum design temperatures of 70–80 °C (Vattenfall, 2011; Van Ginkel et al., 2011; E.ON, 2011b). Only the Longannet project, which use an existing pipeline, indicates a lower maximum design temperature of 30 °C to safeguard cracking (ScottishPower CCS Consortium, 2011):

$$I_{comp} = I_0 * \left(\frac{W_{comp}}{W_{comp,0}} \right)^y * n^{me} \quad (12)$$

where I_{comp} are the investment costs of the compressor including dehydration (M€); I_0 are the base costs (21.9 M€); W_{comp} is the capacity of the compressor (MW_e); $W_{comp,0}$ is the base scale of the compressor (13 MW_e); y is the scaling factor (0.67), n is the number of units in parallel; and me is the multiplication exponent (≈ 0.9).

4.2.2. Energy consumption

It is assumed that the CO₂ inlet conditions for the compressor are slightly above atmospheric pressure (0.11 MPa) and at a temperature of 30 °C. The energy consumption and required capacity for compression is calculated with Eqs. (13) and (14), adapted from Damen et al. (2007) and Kuramochi et al. (2012). It is assumed that the maximum compressor ratio of one stage is 2.04. Hence, six stages are needed for pressurizing to 8 MPa. After the sixth stage, the CO₂ is a liquid and can be pumped further to the required outlet

²³ In literature, installation factors of 2.25–2.82 are mentioned (Woods, 2007). The highest factor is used because CO₂ compressors are complicated equipment due to the large number of stages.

²⁴ In several cases two trains are installed instead of one train (for instance, for the Longannet compressors), while it could be realized with one compressor. This may be due to operational advantages as the plant can more easily operate on half load.

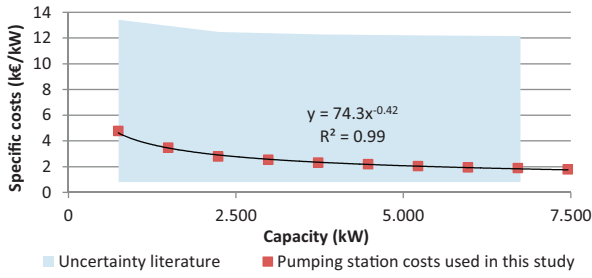


Fig. 5. Specific costs for pumping stations (based on Interliance LLC for California Energy Commission, 2002) with the cost uncertainty range found in literature (Knoope et al., 2013).

pressure:

$$E_{comp} = \sum_{x=1}^{X_{stage}} \frac{Z_x * R * T * \kappa_x}{M * \eta_{iso} * \eta_{mech} * (\kappa_x - 1)} * \left[\left(\frac{P_{2,x}}{P_{1,x}} \right)^{((\kappa_x - 1)/\kappa_x)} - 1 \right] + \frac{P_2 - P_1}{\eta_{pump} * \rho} \quad (13)$$

$$W_{comp} = E_{comp} * m \quad (14)$$

where E_{comp} is the energy consumption of compression (kJ/kg); X_{stage} is the total number of compression stages; x is the compression stage number; Z_x is the compressibility factor of the CO₂ in stage x ²³; R is the Universal gas constant (=8.3145 J/mol/K); T is the inlet temperature compressor stage (303.15 K); κ_x is the specific heat ratio of the CO₂ in stage x ; M is the molecular mass of CO₂ (=44.01 g/mol); η_{iso} is the isentropic efficiency of the compressor (80%); η_{mech} is the mechanical efficiency of the compressor (99%); $P_{2,x}$ is the outlet pressure of compression stage x (MPa); $P_{1,x}$ is the inlet pressure of compression stage x (MPa); P_2 is the outlet pressure of the pump; P_1 is the inlet pressure of the pump (=7.7 MPa); η_{pump} is the efficiency of the pumping equipment (=75%); W_{comp} is the capacity of the compressor (kW_e) and m is the mass flow (kg/s).

4.3. Pumping stations

4.3.1. Capital costs

The design between a water and CO₂ pump is not significantly different if the CO₂ is in the liquid phase. Therefore, the costs of water pumps are used as an approximation for the costs of CO₂ pumps in this study (Mallon and Guijt, 2012). The costs for stand-alone water pumps are given for several capacities in (Interliance LLC for California Energy Commission, 2002). There are relatively strong economies of scale, see Fig. 5 and Eq. (15).

The maximum capacity of a CO₂ pump is 2.0 MW_e (IEA GHG, 2002). For larger capacities, two pump units have to be installed in parallel, each with the same capacity. An additional advantage of installing multiple pumps in parallel is that it can better handle variations in mass flow. Like with the compressors, a train advantage via a multiplication factor is applied (Meerman et al., 2012):

$$I_{pump} = 74.3 * W_{pump}^{0.58} * n^{me} \quad (15)$$

where I_{pump} are the investment costs of pumping stations (€); W_{pump} is the capacity of pumping station per unit (kW_e); n is the number of units in parallel and me is the multiplication exponent (=0.9).

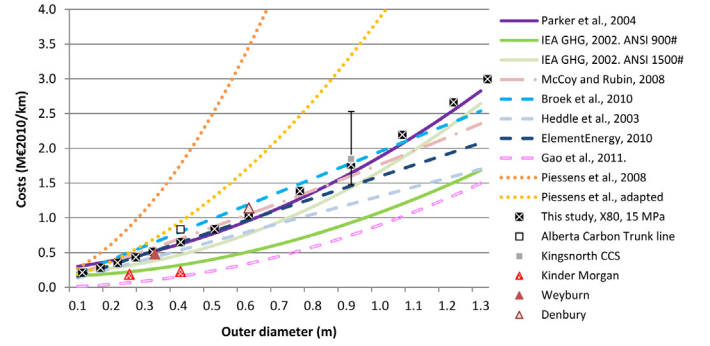


Fig. 6. Comparison of the developed pipeline cost model with cost models given in literature as well as some cost estimations for planned and existing CO₂ pipelines (Knoope et al., 2013).

4.3.2. Energy consumption

The energy requirement and capacity for pumping stations are calculated with Eqs. (16) and (17) (IEA GHG, 2002):

$$E_{pump} = \frac{P_2 - P_1}{\eta_{pump} * \rho} \quad (16)$$

$$W_{pump} = E_{pump} * m \quad (17)$$

where E_{pump} is the energy consumption of pumping (MJ/kg); P_2 is the outlet pressure (MPa); P_1 is the inlet pressure (MPa); η_{pump} is the efficiency of the pumping station (75%); ρ is the density (kg/m³); W_{pump} is the capacity of pumping station (MW_e) and m is the mass flow (kg/s).

5. Results

In Section 5.1, the developed pipeline cost model is compared with cost models available in literature. In Sections 5.2–5.5, the results of the cost minimization process for point-to-point pipelines, pipelines crossing multiple terrains, simple networks and the effect if sources come available in different years are presented and discussed. In Section 5.6, the implications of the system boundaries are discussed and in Section 5.7 a sensitivity analysis is presented.

5.1. Pipeline cost model

In Fig. 6, a comparison of the pipeline cost model developed in this study and cost models in literature is given. It shows that the costs estimated by our model for a pipeline of X80 with a MAOP of 15 MPa are in the upper end of the range predicted by the other costs models. Only the model of Piessens et al., shows larger values, especially for larger diameters.²⁵ Furthermore, the cost estimations of our model are comparable with the cost assessments made for the Alberta Carbon Trunkline, Kingsnorth, Weyburn and Denbury CO₂ pipeline.

In Fig. 7, the required thickness and costs of all steel grades included in this study are given for a pipeline on sparsely populated terrain with a diameter of 0.61 and a MAOP of 3, 15 and 20 MPa. For a MAOP of 3.0 MPa, using higher steel grades do not deliver a cost advantage for a diameter of 0.61 m and for all other diameters, because the minimal thickness should be 1.0% of the diameter. Since the costs of a higher steel grade are higher, the (material) costs increase. For MAOP of 15 and 20 MPa, the thickness

²⁵ The model of Piessens et al. (2008) give higher costs even if the conservative thickness formula of Piessens et al., is adapted to the one used in this study, see Piessens et al., adapted in Fig. 6.

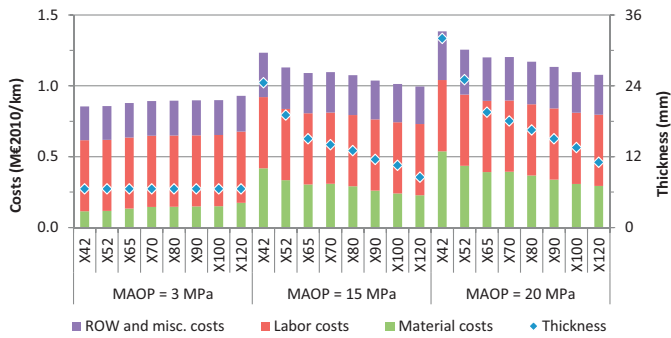


Fig. 7. Cost estimations and thickness required for different steel grades and MAOP on flat sparsely populated terrain for a pipeline with a diameter of 0.61 m.

decreases with higher steel grades and this results in a cost advantage, which is increasing for higher MAOP. For instance, using X120 instead of X80 for a pipeline diameter of 0.61 m gives a capital cost advantage of 7% for a MAOP of 15 MPa and 8% for 20 MPa. If the same pipeline is placed in a populated area or the diameter is enlarged, the cost advantage increases, due to the higher steel requirements. For example, a cost reduction up to 15% is realized in sparsely populated areas and up to 17% in populated areas for a 1.42 m pipeline. For offshore pipelines, there is only a cost advantage of using higher steel grades for large sized pipelines with a high MAOP, due to the requirement that the thickness should be minimal 2.5% of OD_{NPS} .

Notice that for a given mass flow, a higher operation pressure means that the diameter could be decreased. In the supplementary material, a specific case is included, which lead to similar trends as the ones described above.

5.2. Cost minimization of point-to-point pipelines

In Fig. 8, the levelized costs are given for each diameter for a fixed inlet pressure of 12 MPa and the one resulting in the lowest levelized costs is marked. It can clearly be seen that the additional costs of selecting a OD_{NPS} too small are larger than selecting a OD_{NPS} too large.

In Table 3, the results of the cost minimization process are presented for several point-to-point pipelines over one type of terrain. Additional results are given in the supplementary material. Several observations can be made:

- The levelized costs are increasing for decreasing mass flows. For instance, the levelized costs (excluding compression) are 1.8–3.3€/t for transporting 100 kg/s liquid CO₂ over 100 km costs and 1.2–1.8€/t for 200 kg/s. For gaseous CO₂ transport, the

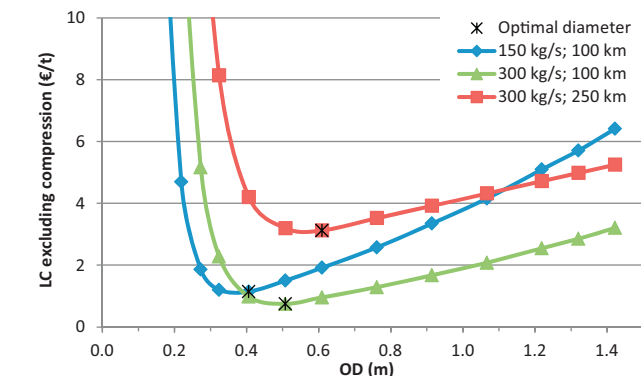


Fig. 8. Optimal diameter for different mass flows and distances on flat, sparsely populated terrain for a fixed inlet pressure of 12 MPa and no pumping stations.

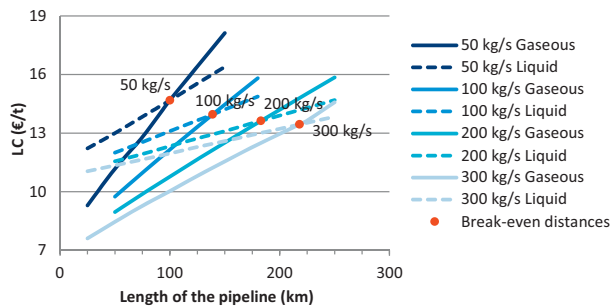
levelized costs are 4.0–6.4€/t and 3.0–3.8€/t for 100 kg/s and 200 kg/s, respectively.

- The optimal pressure drop for liquid CO₂ transport is 15–45 Pa/m for mass flows of 100 kg/s or larger. For smaller mass flows, optimal pressure drops can be up to 100 Pa/m.
- The pipelines through populated terrain have a slightly higher pressure drop than pipelines through sparsely populated terrain because they have a larger wall thickness and consequently a lower inner diameter (compare case number 1; 4; 6 with 7; 8; 9).
- A significant amount of pumping stations are installed along the route, roughly every 50–100 km. This is more cost-effective than increasing the inlet pressure because this means that a pipeline with a larger wall thickness should be installed which would increase the material costs.
- For liquid offshore CO₂ transport, the optimal specific pressure drop is about 10–50 Pa/m, where the low specific pressure drops are for long offshore pipelines transporting large volumes. This can, for instance, be observed by a pipeline transporting 300 kg/s liquid CO₂ (about 9 Mt/y). With a length of 100 km the optimal inlet pressure is 12 MPa and with 340 km the inlet pressure is increased to 22 MPa. For a pipeline length of 350 km, a larger diameter is selected (namely 0.61 m instead of 0.51 m) to limit the inlet pressure (to 14 MPa) and the specific pressure drop (from 40 to 14 Pa/m), see case number 11–13. Similar trends can be seen by different mass flows, and the maximum inlet pressure is often about 25 MPa. Hence, it is more cost-effective to increase the diameter rather than to increase the inlet pressure to very high pressures for long offshore pipelines.
- For gaseous CO₂ transport, relatively large diameters are selected due to the low density of CO₂. Consequently, the specific pressure drop is lower than for liquid transport, namely 5–10 Pa/m (see case number 14–16).
- The optimal velocity is about 1–2 m/s for pipelines transporting liquid CO₂ and about 5–15 m/s for pipelines transporting gaseous CO₂.
- For onshore pipelines transporting liquid CO₂, X80 is always selected as optimal steel grade in the short term, X100 in the medium term and X120 in the long term (case number 1–3). The levelized cost are decreasing with 3–12% in the long term, where the largest cost decrease is realized with larger diameters. Offshore pipelines have lower optimal steel grades, especially for larger diameters, because of the 2.5% thickness requirement. However, for pipelines with high inlet pressures and small diameters (so long distances for small mass flows) higher steel grades are interesting.
- Gaseous transport can be cost-effective compared to liquid transport because the higher specific pipeline costs for gaseous CO₂ transport are compensated by savings in the compression costs. For instance, with a mass flow of 100 kg/s (about 3.2 Mt/y), the breakeven distance where gaseous and liquid CO₂ transport have equal costs is 139 km for pipelines over sparsely populated terrain, 146 km for pipelines over populated terrain and 94 km for offshore pipelines. Fig. 9 shows that the break-even distance increases with increasing mass flows, until the largest NPS included in the model is too small for the gaseous mass flow.²⁶ The reason for this is that the pipeline costs have larger economies of scale than the compressor costs and consequently the break-even distance where the additional pipeline costs equal the savings in compression costs for gaseous CO₂ transport is increasing with increasing mass flows. Note that this comparison does not take

²⁶ For a pipeline larger than 100 km on sparsely populated terrain and mass flow rates larger than 650 kg/s, the largest available NPS in the model ($OD = 1.42$ m) is not large enough.

Table 3Results for the cost minimization process for pure CO₂. The outlet pressure is fixed on 8 MPa for liquid transport and on 1.5 MPa for gaseous transport.

No.	Time frame ^a	Terrain ^b	Mass flow (kg/s)	Length (km)	OD (m)	P_{inlet} (MPa)	N_{pumps}	L_{pump} (km)	LC_{ALL}^c (€/t)	LC_{trans}^c (€/t)	ΔP_{act} (Pa/m)	Steel grade	Phase
1	Short	S.Pop	100	100	0.32	13	0	114	13.1	1.91	44	X80	Liq.
2	Mid	S.Pop	100	100	0.32	12	1	96	13.0	1.87	42	X100	Liq.
3	Long	S.Pop	100	100	0.32	12	1	99	13.0	1.84	40	X120	Liq.
4	Short	S.Pop	150	100	0.41	11	0	102	12.7	1.57	29	X80	Liq.
5	Short	S.Pop	250	100	0.51	11	0	118	12.2	1.20	25	X80	Liq.
6	Short	S.Pop	500	100	0.61	10	1	52	11.6	0.84	38	X80	Liq.
7	Short	Popu	100	100	0.32	13	0	105	13.2	2.05	48	X80	Liq.
8	Short	Popu	150	100	0.41	11	1	96	12.9	1.71	31	X80	Liq.
9	Short	Popu	500	100	0.61	10	2	49	11.7	0.95	41	X80	Liq.
10	Short	Offsh	100	100	0.32	13	n.a.	n.a.	14.4	3.25	43	X65	Liq.
11	Short	Offsh	300	100	0.51	12	n.a.	n.a.	12.4	1.43	38	X52	Liq.
12	Short	Offsh	300	340	0.51	22	n.a.	n.a.	15.7	4.35	40	X80	Liq.
13	Short	Offsh	300	350	0.61	14	n.a.	n.a.	15.9	4.91	14	X65	Liq.
14	S/M/L	S.Pop	100	100	0.76	2.5	n.a.	n.a.	12.2	4.00	9.6	X42	Gas.
15	S/M/L	Popu	100	100	0.76	2.5	n.a.	n.a.	12.2	4.02	9.6	X52	Gas.
16	S/M/L	Offsh	100	100	0.76	2.6	n.a.	n.a.	14.6	6.43	10	X42	Gas.

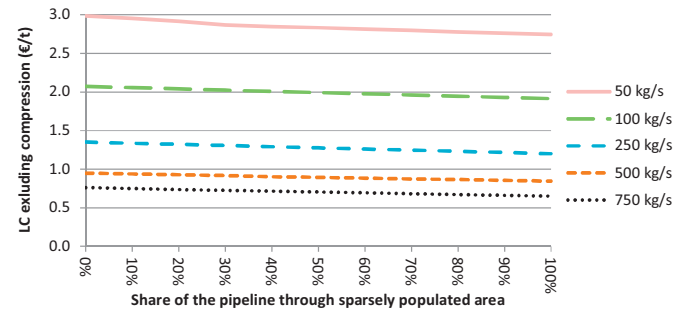
^a S/M/L stands for short, medium and long term.^b S.Pop, popu and offsh refers to sparsely populated, populated terrain and offshore pipelines, respectively.^c LC_{ALL} refers to the levelized costs associated with initial compression, pipeline and pumping stations, while LC_{trans} refer to the costs associated with only pipeline and pumping stations.**Fig. 9.** Breakeven distance between liquid and gaseous CO₂ transport for pipelines over sparsely populated terrain.

into account storage field requirements. When the inlet pressure to the sink has to be 8 MPa or higher,²⁷ then it would be more cost-effective to compress it at the plant and transport it as a liquid rather than transporting it as a gas and compress it (further) at the storage location, see also Section 5.6.

5.3. Pipeline crossing different types of terrain

In Fig. 10, the levelized costs are depicted for a pipeline of 100 km, where different shares of the pipeline transverse through sparsely populated areas. The levelized costs decrease if a larger share of the pipeline passes through sparsely populated instead of populated terrain. Changing the terrain from sparsely populated to populated terrain increase the levelized cost with 8% (for 100 kg/s) up to 18% (for 750 kg/s) for a 100 km pipeline. With longer distances, the cost increase is slightly less, because pumping stations have (assumed) similar costs on sparsely populated and populated terrain.

In Fig. 10, the consequence of only higher material costs are included for pipelines through populated areas. However, it is likely

**Fig. 10.** Levelized costs (excluding compression) for a pipeline of 100 km passes partly through sparsely populated and partly through populated areas for different mass flows in the short term.

that the ROW cost would also increase if the pipeline is installed on populated instead of sparsely populated area. Unfortunately, no data is available on the exact increase. By assuming that the ROW costs would double in populated areas, the levelized costs excluding compression would increase with 20% (for mass flows of 500 kg/s) to 31% (for mass flows of 50 kg/s). With a tripling of the ROW costs, the levelized costs would increase with 27% (for mass flows of 500 kg/s) to 53% (for mass flows of 50 kg/s). In the rest of the article, ROW costs are assumed to be equal on populated and sparsely populated terrain, due to the lack of better data.

In Table 4, the results are presented of several pipelines transporting CO₂ from onshore sources to offshore storage locations. In all cases a pumping station is installed just before the pipeline transits offshore. This is more cost-effective than increase the pressure immediately at the capture plant because this leads to higher material costs.

5.4. Simple network approach

In this paragraph, the configuration of a simple network is optimized and it is evaluated which of the four configuration stated in the methodology appears the most cost-effective. In literature, several outlines of networks for CCS are proposed and analyzed. In this study, we used an outline of a simple network proposed by ZEP, which is a consortium consisting of stakeholders of CCS. The simple network contains of two 10 km long feeders of 1330 t/h (≈ 370 kg/s), one trunkline of 180, 500, 750 or 1500 km transporting 2660 t/h (≈ 740 kg/s) and two distribution pipelines of 1330 t/h and

²⁷ The inlet pressure to the sink has to match the reservoir pressure corrected for the head and the frictional losses in the well. Although there is some operational freedom by selecting the right diameter for the well and playing with the temperature of the CO₂ and the mass flow, the reservoir pressure determines the inlet pressure. Typically, depleted gas fields have in the beginning quite a low reservoir pressure (about 1–5 MPa) while aquifers and (depleted) oil fields have a higher reservoir pressure (about 8–15 MPa). Hence, for depleted gas fields gaseous CO₂ transport may be interesting.

Table 4

Results of the cost minimization process for a pipeline partly onshore and partly offshore. The outlet pressure of the entire pipeline is fixed on 8 MPa.

No.	Time frame	Mass flow (kg/s)	Onshore section						Offshore section					N_{pumps}^a	LC_{All}^b (€/t)	$LC_{Transport}^b$ (€/t)
			Terrain	Length (km)	OD (m)	P_{inlet} (MPa)	Steel grade	P_{outlet} (MPa)	Terrain	Length (km)	OD (m)	P_{inlet} (MPa)	Steel grade			
17	Short	250	Spar. popu.	100	0.51	11	X80	8	Offshore	100	0.51	11	X52	0+1	14.2	3.16
18	Mid	250	Spar. popu.	100	0.51	11	X100	8	Offshore	100	0.51	11	X52	0+1	14.1	3.11
19	Long	250	Spar. popu.	100	0.51	11	X120	8	Offshore	100	0.51	11	X52	0+1	14.1	3.09
20	Short	100	Spar. popu.	250	0.32	11	X80	8	Offshore	750	0.41	19	X80	3+1	36.3	25.2
21	Short	250	Spar. popu.	250	0.51	9	X80	8	Offshore	750	0.61	16	X80	6+1	27.1	16.1
22	Short	250	Spar. popu.	250	0.51	9	X80	8	Offshore	500	0.51	22	X80	6+1	22.5	11.5
23	Short	250	Spar. popu.	250	0.51	9	X80	8	Offshore	250	0.51	15	X65	6+1	18.3	7.40
24	Short	250	Spar. popu.	250	0.51	9	X80	8	Offshore	100	0.51	11	X52	6+1	16.2	5.29
25	Short	100	Spar. popu.	100	0.32	13	X80	8	Offshore	100	0.32	13	X65	0+1	16.7	5.48
26	Short	500	Spar. popu.	100	0.61	10	X80	8	Offshore	100	0.61	12	X65	1+1	12.9	2.16
27	Short	750	Spar. popu.	100	0.76	10	X80	9	Offshore	100	0.76	11	X52	1+1	12.5	1.72
28	Short	250	Populated	100	0.51	10	X80	8	Offshore	100	0.51	11	X52	1+1	14.3	3.30

^a The first number refers to the number of pumping stations installed on the onshore track, the second number refers if a pumping station is installed just before the pipeline goes offshore.^b LC_{All} refers to the levelized costs associated with initial compression, pipeline and pumping stations, while LC_{trans} refer to the costs associated with only pipeline and pumping stations.**Table 5**

Comparison of diameters and costs of an onshore network from ZEP and this model.

No.	Length trunkline (km)	Location trunkline	Mass flow (t/h)	Length feeders & distribution lines (km)	ZEP			This model ^a							
					OD trunkline (m)	OD feeders (m)	LC^b (€/t)	OD trunkline (m)	OD feeders (m)	P_{inlet} (MPa)	P_{outlet} (MPa) ^c	N_{pumps}	Network option ^d	LC_{All}^e (€/t)	LC_{trans}^e (€/t)
29	180	Offshore	2 × 1330	10	0.56	0.51	3.4	0.76	0.61	14	8	0	III	13.3	1.97
30	500	Offshore	2 × 1330	10	0.66	0.51	6.0	0.76	0.61	12	6	0	III	13.2	1.92
								0.91	0.61	14	8	0	III	17.1	5.77
31	750	Offshore	2 × 1330	10	0.66	0.51	8.2	0.76	0.61	21	6	0	III	17.0	5.23
								0.91	0.61	17	8	0	III	20.3	8.81
32	1500	Offshore	2 × 1330	10	0.76	0.51	16.3	0.91	0.61	15	6	0	III	20.0	8.57
								1.07	0.61	16	8	0	III	32.9	21.4
33	180	Onshore	2 × 1330	10	0.81	0.51	1.5	1.07	0.61	14	6	0	III	31.8	20.5
								0.76	0.61	9	8	4	III	13.1	1.96
34	500	Onshore	2 × 1330	10	0.81	0.51	3.7	0.76	0.61	10	8	6	III	15.8	4.64
35	750	Onshore	2 × 1330	10	0.81	0.51	5.3	0.76	0.61	10	8	10	III	17.9	6.74

^a The onshore pipeline costs given in the ZEP rapport are specific for flat topography, simple soil conditions and unobstructed ROW. Therefore, the results of this model for onshore pipelines are based on sparsely populated terrain and steel grades available in the short term. Furthermore, in the ZEP report it is assumed that the yearly operation time is 7500 h. This number is also used for calculating the costs and configurations in this table.^b The levelized cost of the ZEP study includes the pipeline itself and intermediate pumping stations. The cost for compression till 11 MPa and dehydration of the CO₂ are included in the capture costs and not in the transportation costs. The number of pumping stations included in the configuration is not mentioned.^c In this study, the standard outlet pressure is 8 MPa. In the ZEP study, the outlet pressure is 8 MPa for onshore pipelines and 6 MPa for offshore pipelines. Hence, the results of the model are calculated for 8 as well as for 6 MPa for the offshore cases.^d As described in the methodology: I = gaseous transport in the feeders as well as in the trunkline and distribution pipelines; II = gaseous transport in the feeders and liquid transport at the trunkline and distribution pipelines; III = liquid transport in the entire network, where the streams flow immediately from the feeders into the trunkline without a pumping station; IV = liquid transport in the entire network, having a pumping station installed just before at the inlet of the trunkline.^e LC_{All} refers to the levelized costs associated with initial compression, pipeline and pumping stations, while LC_{trans} refer to the costs associated with only pipeline and pumping stations.

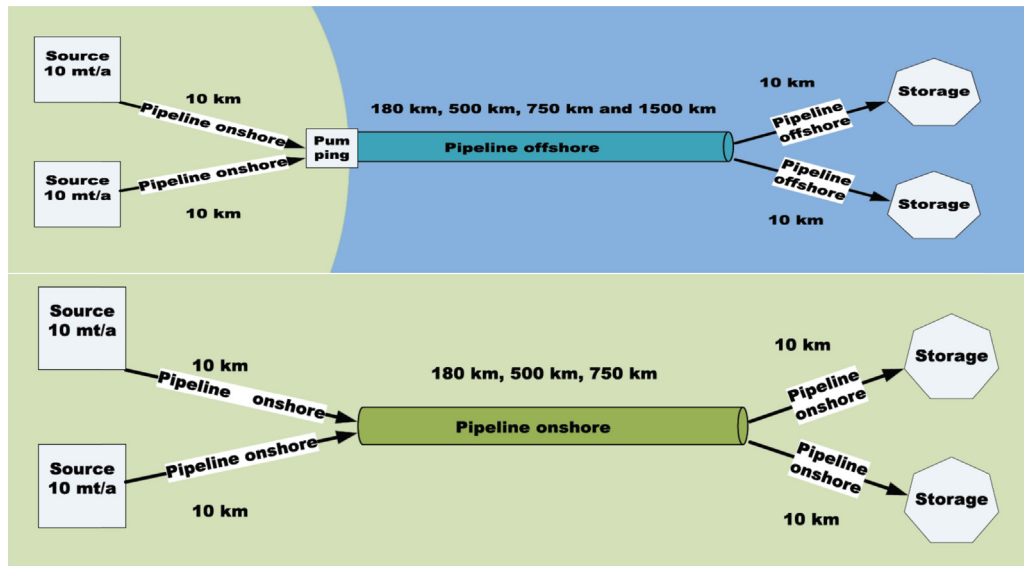


Fig. 11. Overview of a simple complete onshore (above) and partly offshore (below) network with one trunkline of 740 kg/s, two onshore feeders of 370 kg/s and two offshore distribution pipelines of 370 kg/s (ZEP, 2010).

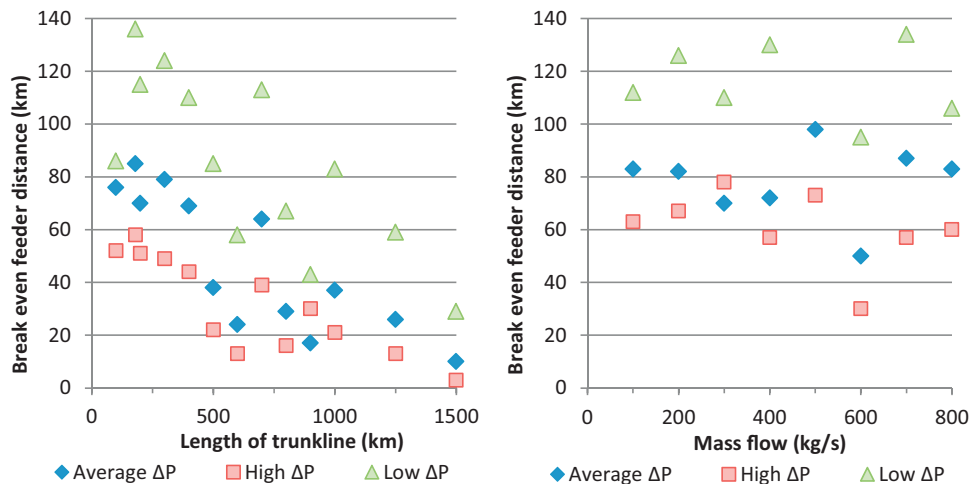


Fig. 12. Breakeven feeder distance between option III (no pumping station before trunkline) and IV (pumping station before trunkline) for an offshore trunkline transporting liquid CO₂ with two distribution lines of 10 km with an average (30 Pa/m for liquid CO₂), high (+50%) or low (−50%) design pressure drop. On the left, the distance of the trunkline is varied and the mass flow is kept constant at 740 kg/s (=20 Mt/y with an operation time of 7500 h). On the right, the length of the trunkline is fixed at 180 km and the mass flows (with an operation time of 7500 h) is varied.

10 km, see Fig. 11 (ZEP, 2010). The model developed in this study is used to optimize these networks and the results of this are compared with the configurations given by ZEP. The results of the ZEP study as well as for this study are listed in Table 5.

It is important to point out several differences between the configuration of ZEP and the model developed in this study:

- In the ZEP study, a fixed inlet (10 MPa onshore and 20 MPa offshore) and outlet pressures (8 MPa onshore and 6 MPa offshore) is assumed. Consequently, for an offshore trunkline of 180 km a very high specific pressure drop can be realized, leading to a relatively small diameter (case 29). However, it would be more cost-effective to increase the diameter and decrease the inlet pressures.
- In the ZEP study, it is assumed that always a pumping station is installed before the pipeline goes offshore and no pumping station before the trunkline for onshore situations. However, the results of this study show that compression at the plant is a more cost-effective alternative with short feeders of 10 km for 370 kg/s

for offshore pipelines. Additionally, installing a pumping station just before the onshore trunkline may have a costs advantage for long feeders and trunklines.

For offshore trunklines, option IV (see Section 3.3.1) becomes the most cost-effective option if the distance between the feeders and the trunkline is long. For instance, for an offshore trunkline of 180 km, transporting 740 kg/s (about 20 Mt/y) and distribution pipelines of 10 km, option III is the best option for feeders up to 84 km on sparsely populated terrain and up to 78 km for populated terrain. For longer feeders, option IV becomes economically the best alternative. If the specific design pressure in the feeders and the distribution lines is increased, the MAOP of the trunklines has to be increased already with shorter feeder distances. Hence, the average breakeven distances decrease with higher specific pressure drops, see Fig. 12. In Fig. 12, it can also be assessed that for longer trunklines the breakeven distance is decreasing. The reason for this is that the total steel costs for option III increase considerably due to the higher MAOP of the entire system. Additionally, there is no relation

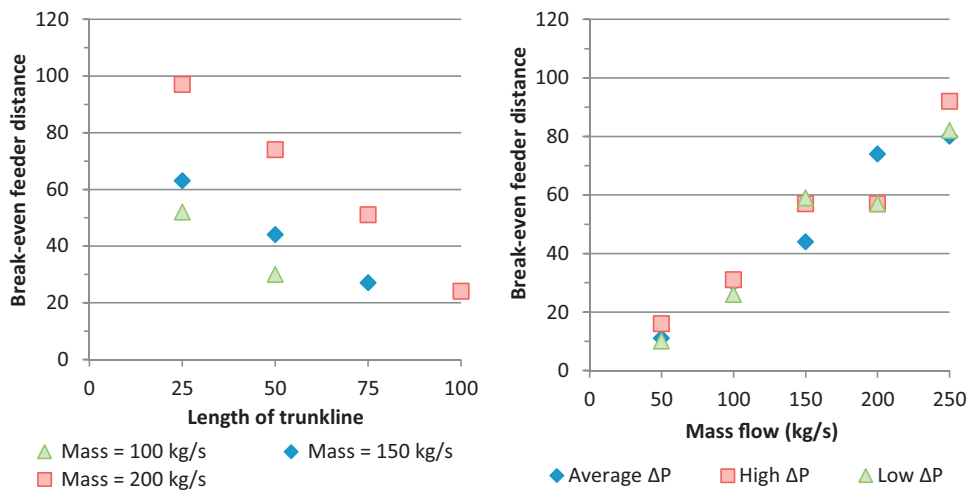


Fig. 13. Breakeven feeder distance between gaseous transport in whole the network (option I) and liquid CO₂ transport in part of or whole the network (option II, III or IV) with a trunkline and distribution lines of 10 km on sparsely populated terrain. On the left, the distance of the trunkline is varied for three different mass flows (100, 150 and 300 kg/s with an operation time of 7500 h). (Gaseous transport in whole the network (option I) is never the most cost-effective for a mass flow of 100 kg/s in combination with a trunkline of 75 or longer, a mass flow of 150 kg/s in combination with a trunkline of 100 km or longer and a mass flow of 200 kg/s in combination with a trunkline of 125 km or longer.) On the right, the length of the trunkline is fixed at 50 km and the mass flows (with an operation time of 7500 h) is varied for three different specific design pressure drops in the feeders and distribution lines.

between mass flow in the trunkline and the break-even distance, see Fig. 12. The variation of the breakeven distances is caused by the limit number of NPS available and the rounding of the thickness and MAOP.

For onshore pipelines, the situation is less straight forward because the length of the feeder has consequences for the operation pressure of the trunkline in network option III. Hence, pumping stations along the pipeline can pump the CO₂ to a higher pressure and one or more pumping stations may be eliminated. Consequently, the levelized costs of option III have a “jumpy” character and multiple breakeven points can exist between network option III and option IV.

For networks with short trunklines and low mass flow rates, gaseous CO₂ transport in whole the network (option I) can be the most cost-effective. In Fig. 13, the breakeven distance between gaseous transport in whole the network (option I) and liquid transport in part of or whole the network (option II, III or IV) can be seen for different trunkline lengths, various mass flows and three different design pressure drops. It is clear that the breakeven feeder distance is decreasing with increasing trunkline lengths until liquid CO₂ transport in the trunkline is the most cost-effective option regardless the feeder distance. Furthermore, the breakeven feeder distance is increasing with increasing mass flows.

Option II, where the CO₂ is transported as a gas through the feeders and compressed to a liquid just before the trunkline, is in a few cases the most cost-effective. For instance, option II gives the lowest levelized costs for a mass flow of 2×75 kg/s, a trunkline of 200 km, feeders and distribution lines of 10 km on sparsely populated terrain, namely 16.0€/t compared to 19.9€/t, 16.2 and 16.4€/t for option I, II and IV, respectively. The main reason for this is that the overall compression capacity required for this option (46 MW) is slightly lower than for complete liquid CO₂ transport (50 MW), due to the lower pressure drop in the feeders. However, this is mainly the result of the assumption that the specific pressure drop is lower for gaseous than for liquid CO₂ transport.

5.5. Timing aspects

In the previous paragraph, it was assumed that two sources are transporting CO₂ from the same moment onwards. However, in reality it is likely that one source wants to transport from, for

instance, 2020 onwards while the second source starts from 2025 or 2030.

In Table 6, the configurations and levelized costs are calculated for several combinations of distances between the sources, distances of the trunkline and mass flows of two different sources. Several observations can be made:

- It appears more cost-effective to combine the sources in a trunkline if the sink is further away from the sources (case 36–37). Consequently, the sources can be further apart from each other.
- If the sources have unequal mass flows, it is often cost-effective to construct a trunkline, if the source which come available first has the largest mass flow (case 38–41).
- Two point-to-point-pipelines of 100 km have in most cases slightly lower levelized costs than a trunkline of 100 km for two equally sized sources separated 10 km from each other coming available 10 years after each other (case 42–45). Hence, the additional costs of over-dimensioning a pipeline are almost similar to the costs of constructing a complete new point-to-pipeline in 2030 due to discounting. This effect will become stronger if the time difference between the availability of the sources is enlarged. For similar distances and mass flows, but now with a time difference of 5 years, the trunkline configuration results in all cases (48–51) in slightly lower levelized costs than the point-to-point pipeline configuration. The differences in LC between the trunkline and point-to-point pipeline configuration are however small, especially considering the uncertainties in the different cost models. Nevertheless, it can be concluded that a trunkline configuration becomes less cost-effective with increasing time delays and two point-to-point pipelines are preferred after a time delay of about 5–10 years.
- With a time difference of 5 years, the sources can be further apart from each other and still be cost-effectively combined in a trunkline (compare 46–47 with 52–53).

If the trunkline option is the most cost-effective approach and the second source is not coming available, then the levelized costs will increase to above the level of the point-to-point pipeline approach. For instance, for a trunkline length of 100 km, feeders of 10 km on sparsely populated terrain and two mass flows of 100 kg/s coming available 5 years after each other, the levelized costs of

Table 6

Network optimization for two sources which become available in different time frames, where the first source becomes available in 2020.

No.	Length trunk (km)	Length feeder (km)	Distance source II -sink (km)	Source II ^a availability	Terrain ^b	Mass flow I (kg/s)	Mass flow II (kg/s)	Inlet pressure ^c			N_{pumps} ^c			Diameter ^c			Levelized costs (€/t CO ₂) ^{c,d}			
								Tr.	PtP I	PtP II	Tr.	PtP I	PtP II	Tr.	PtP I	PtP II	LC_{ALL} PtP	LC_{trans} PtP	LC_{ALL} trunk	LC_{trans} trunk
36	200	10	200	2030	S.Pop	150	150	12	11	11	1	1	1	0.51	0.41	0.41	14.6	3.45	14.5	3.36
37	500	10	500	2030	S.Pop	150	150	11	10	10	6	7	6	0.51	0.41	0.41	20.1	9.00	19.6	8.52
38	200	10	200	2030	S.Pop	200	100	12	12	11	1	2	2	0.51	0.41	0.32	14.1	3.16	13.9	2.93
39	200	10	200	2030	S.Pop	100	200	12	11	12	1	2	2	0.51	0.32	0.41	14.7	3.68	15.0	3.95
40	200	10	200	2030	S.Pop	50	100	11	13	11	1	3	2	0.41	0.22	0.32	16.9	5.53	17.3	5.95
41	200	10	200	2030	S.Pop	100	50	11	11	14	1	2	2	0.41	0.32	0.22	15.8	4.65	15.5	4.39
42	100	10	100	2030	S.Pop	50	50	13	18	17	0	0	0	0.32	0.22	0.22	14.7	2.84	14.5	2.97
43	100	10	100	2030	S.Pop	100	100	14	13	12	0	0	1	0.41	0.32	0.32	13.1	1.98	13.0	1.92
44	100	10	100	2030	S.Pop	200	200	10	14	13	1	0	1	0.61	0.41	0.41	12.3	1.27	12.4	1.49
45	100	10	100	2030	S.Pop	500	500	9	10	12	1	1	0	0.91	0.61	0.61	11.6	0.84	11.7	0.98
46	200	25	200	2030	S.Pop	100	100	12	11	11	2	2	2	0.41	0.32	0.32	15.4	4.30	15.2	4.13
47	200	50	200	2030	S.Pop	100	100	12	11	11	2	2	2	0.41	0.32	0.32	15.4	4.30	15.4	4.31
48	100	10	100	2025	S.Pop	50	50	13	18	18	0	0	0	0.32	0.22	0.22	14.7	2.85	14.2	2.59
49	100	10	100	2025	S.Pop	100	100	14	13	13	0	0	0	0.41	0.32	0.32	13.1	1.99	12.8	1.68
50	100	10	100	2025	S.Pop	200	200	10	14	14	1	0	0	0.61	0.41	0.41	12.3	1.28	12.2	1.35
51	100	10	100	2025	S.Pop	500	500	9	10	10	1	1	1	0.91	0.61	0.61	11.6	0.87	11.5	0.87
52	200	50	200	2025	S.Pop	100	100	12	11	11	2	2	2	0.41	0.32	0.32	15.4	4.33	15.2	4.05
53	200	75	200	2025	S.Pop	100	100	12	11	11	2	2	2	0.41	0.32	0.32	15.4	4.33	15.5	4.33

^a In 2025, the same steel grades are available as in the short term.^b S.Pop, popu and offsh refers to sparsely populated, populated terrain and offshore pipelines, respectively.^c The abbreviations Tr, PtP I, and PtP II refer to trunkline, point-to-point pipeline from source I, and point-to-point pipeline from source II to the sink, respectively.^d LC_{ALL} refers to the levelized costs associated with initial compression, pipeline and pumping stations, while LC_{trans} refer to the costs associated with only pipeline and pumping stations. The network approach (point-to-point pipelines or trunkline) which result in the lowest LC_{ALL} , is italic and bold.

Table 7
Sensitivity results of one case for transporting 500 kg/s CO₂ over 100 km sparsely populated terrain. A complete overview of all cases can be found in the supplementary material.

		OD (m)	P_{inlet} (MPa)	N_{pumps}	LC_{ALL} (€/t CO ₂)	LC_{trans} (€/t CO ₂)	Pressure drop (Pa/m)	L_{pump} (km)
Base case		0.61	10	1	11.6	0.84	38	52
Electricity costs	–50%	0.61	10	1	7.02	0.81	38	52
	+50%	0.76	9	1	16.2	0.95	12	84
Steel costs	–50%	0.61	12	0	11.5	0.64	40	101
	+50%	0.61	10	1	11.7	0.94	38	52
Labor costs	–50%	0.76	9	1	11.4	0.66	12	84
	+50%	0.61	10	1	11.9	1.08	38	52
	+100%	0.61	10	1	12.1	1.31	38	52
Interest rate	–50%	0.76	9	1	10.9	0.57	12	84
	+50%	0.61	10	1	12.4	1.16	38	52

LC_{ALL} refers to the levelized costs associated with initial compression, pipeline and pumping stations, while LC_{trans} refer to the costs associated with only pipeline and pumping stations.

combining the two sources would be 12.3€/t (1.7€/t excluding compression) but if the second source would not become available they will increase to 13.3€/t (2.6€/t excluding compression) while the levelized costs for a separate point-to-point pipeline would be 12.5€/t (2.0€/t excluding compression).

5.6. Implications of the system boundaries

In this study, gaseous and liquid CO₂ transport are compared and in several cases gaseous transport resulted in lower levelized costs than liquid CO₂ transport. However, the system boundaries have been placed before the injection well. If CO₂ storage requirements are taken into account the preference between gaseous and liquid CO₂ transport could change. To analyze this, two different cases are explored. Firstly, a case with a required injection pressure of 8 MPa, which represents injection in an aquifer. Secondly, a case representing injection in a depleted gas field, where the required injection pressure increase linearly in three years from 1.5 to 8 MPa and then remains on 8 MPa for the rest of the lifetime.²⁸ For this analysis, it is assumed that a stand-alone compressor is as expensive as the initial compressor near the capture plant.

For the first case, the gaseous CO₂ has to be compressed at the storage location from the original pressure of 1.5 MPa to the required injection pressure of 8 MPa. This results in costs of 5.1€/t for 100 kg/s CO₂. These additional costs are enough to cancel out the cost advantage of gaseous compared to liquid CO₂ transport, which was 0.9€/t for 100 km and 2.3€/t for a 50 km pipeline over sparsely populated land. Similar results are found for other mass flows and pipeline length. Thus, if the CO₂ has to be injected with a pressure of 8 MPa, it is more cost-effective to transport CO₂ as a liquid than as a gas.

For the second case, a compressor has to be installed in the second year which is able to compress the gaseous flow to 3.7 MPa in the second year, to 5.8 MPa in the third year and to 8 MPa from year four onwards. For a mass flow of 100 kg/s, a compressor of 11 MW_e is needed. This increases the levelized costs with 3.5€/t and results in total levelized costs of 13.2 and 15.6€/t for 50 and 100 km pipeline on sparsely populated terrain. Note that in the first few years, liquid CO₂ has to be decompressed so that it can be injected in the depleted gas field. However, this will lead to a large drop in temperature and injection of cold CO₂ can lead to hydrate formation in the well and to thermal cracking of the surrounding rocks in the well formation. Therefore, an electric heater is required to increase the injection temperature to 0 °C, which is estimated to be high enough to avoid hydrate formation and thermal

²⁸ In practice, the required injection pressure will further increase. However, this will be similar for the gaseous and liquid CO₂ case and does not change the preference. Therefore, it is not included.

cracking (E.ON, 2011d). A heater of 20 MW is installed and used for two years.²⁹ With estimated heater costs of 1 M€/MW (E.ON, 2011c,d)³⁰ and O&M costs of 4% of the capital costs, the heater increases the levelized costs with 1.6€/t for a mass flow of 100 kg/s. Combining these costs with the initial compression and transportation costs, leads to levelized costs for liquid CO₂ transport of 13.4 and 14.1€/t for 100 kg/s over 50 and 100 km sparsely populated terrain, respectively. If these costs are compared with the levelized costs for gaseous CO₂ transport, the costs for liquid CO₂ transport are lower for the 100 km but higher for 50 km case.

To conclude, for storage reservoirs with a high reservoir pressure, liquid CO₂ transport appear the best option. For depleted gas fields with a low reservoir pressure, the cost effectiveness of liquid or gaseous CO₂ transport depends on the source–sink distance, mass flow and on how quickly the required injection pressure increase over time. Gaseous CO₂ transport seems to be especially interesting for small CO₂ mass flows on a close distance from a depleted gas field with a low reservoir pressure.

5.7. Sensitivity analysis

A sensitivity analysis is conducted to assess the robustness of the optimal configuration and the impact on the levelized costs for point-to-point pipelines. Four of the main economic input parameters are varied, namely the interest rate, electricity, steel and labor costs (Table 1). The sensitivity analysis is conducted for different mass flows, lengths, time frames and terrain types. In total about 50 cases are analyzed and 20 of them focus on pipelines transporting liquid CO₂ over sparsely populated terrain in the short term. The results of one case are given in Table 7, but results of all runs are reported in the supplementary material.

The average consequences of a 50% increase in interest rate, steel price, electricity or labor costs on the levelized costs including and excluding compression are shown in Fig. 14. The results indicate that the total levelized costs are mostly influenced by the electricity

²⁹ Isobaric heating and adiabatically expansion is assumed by calculating the energy required for heating up the CO₂. In the first year, the CO₂ is injected with a pressure of 1.5 MPa and temperature of 0 °C and delivered at 8.0 MPa and temperature of 15 °C. To ensure similar enthalpy levels before and after expansion, the CO₂ has to be first heated up to 72 °C. Because the specific heat under isobaric conditions varies quite a lot over this temperature range (Span and Wagner, 1996), a weighted average of the heat capacity is used. Subsequently, the required heater capacity is calculated by assuming a heater efficiency of 100%, ($m \cdot C_p \cdot \Delta T / \eta = 100 \text{ kg/s} \cdot 3.44 \text{ kJ/K} \cdot (72 - 15) / 100\% = 19.6 \text{ MW}$).

³⁰ In literature, costs for large electric heaters are not available, as far as the authors know. In the FEED study of E.ON, 4 heaters of 1.05 MW are included (E.ON, 2011d). The costs for this are probably included in the category ‘well interface’, together with the choke valve, safety valve and metering equipment. In total, this category costs 4.66 M€₂₀₁₀ (E.ON, 2011c). Based on this information the costs of the heater are estimated at 1 M€/MW.

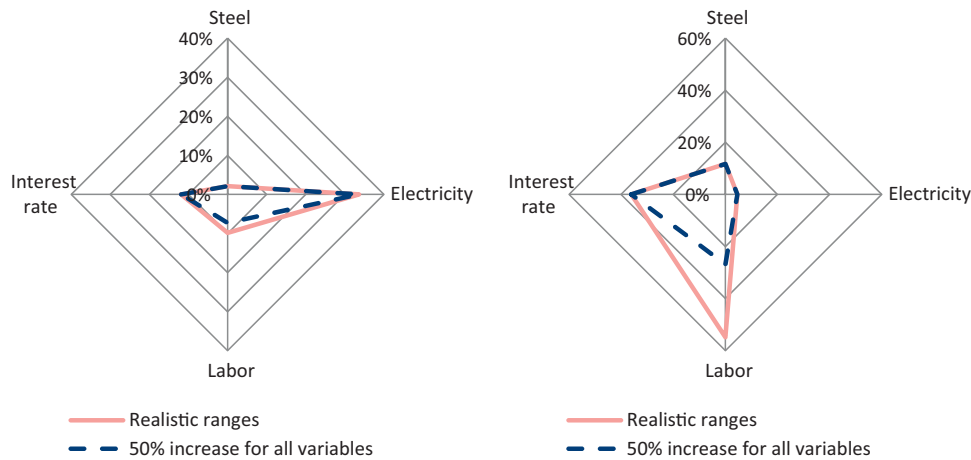


Fig. 14. The average consequences of an increase in interest rate, steel, electricity, and labor costs on the levelized costs, for CO₂ transport on flat sparsely populated terrain, including (left) and excluding compression costs (right).

costs due to the high compression energy. If the compression costs are excluded, the interest rate and labor have the largest impact followed by steel and electricity costs. Similar trends can be seen when the parameters are all increased by 100%.

However, the labor costs are more uncertain than the other three variables. Hence, a 100% increase in the labor costs is assumed to be as likely as a 50% increase in the interest rate, electricity and steel costs. If these realistic ranges are taken into account, the levelized costs including compression are still mostly influenced by the electricity costs, see Fig. 14. However, the levelized costs without compression are now most influenced by the uncertainty in labor costs.

Besides influencing the levelized costs, varying a parameter, e.g. with 50% or 100% can also lead to a different optimal pipeline configuration (see Table 7 and supplementary material). Overall, changes on the steel price has the largest impact on the optimal inlet pressure. For instance, at a lower steel price, it is more cost-effective to invest in a pipeline with a thicker wall and install less pumping stations along the route.

In contrast, it is often cost-effective to install a larger diameter to limit the specific pressure drop if the interest rate or labor costs decrease or the electricity costs increase. The initial higher investment costs are compensated by the lower energy consumption during the lifetime. In most cases, the optimal configuration does not change with a 50% increase in the interest rate or labor costs. However, with a 100% increase in the labor costs or with a 50% reduction in the electricity costs, the selection of a smaller diameter is cost-effective in several cases.

Overall, offshore cases appear less sensitive to changes in the interest rate, electricity, steel and labor costs than onshore cases, because the option to add or eliminate pumping stations is not available. Hence, the cost minimization is only driven by the NPS available. This argument is also valid for gaseous CO₂ transport.

Additionally, it has also been examined whether a different interest rate could affect the cost-effectiveness of a trunkline versus point-to-point pipelines when two sources come available in different time frames. For this, 35 cases were run with a time difference of 5 or 10 years between the availability of the first and second CO₂ source (see the supplementary material for the results per case). If the interest rate is increased to 15%, the cost-effectiveness of the trunkline decreases compared to the point-to-point pipeline configuration. The reason for this is the strong discount effect of, in particular, the second point-to-point pipeline. Consequently, even with a time difference of only five years between two equally sized sources 10 km apart from each other, several cases prefer two

point-to-point pipelines to a trunkline. The opposite is also true, a lower interest rate has a positive effect on the trunkline configuration. With respect to configuration changes, similar trends were observed as described above for the trunkline as well as for the point-to-point pipeline configurations.

6. Conclusion

This study had two main aims. Firstly, to develop a new cost model for CO₂ pipeline transport related to the physical properties of CO₂, and secondly to develop a cost minimization model to determine the optimal configuration for point-to-point pipelines as well as for simple networks.

The new pipeline cost model explicitly takes into account differences in operation pressure and materials. The results indicate that low steel grades, such as X42 and X52, lead to the lowest capital costs for pipelines transporting gaseous CO₂, while higher steel grades lead to the lowest capital costs for onshore pipelines transporting liquid CO₂. For instance, for an onshore pipeline with a diameter of 0.61 m and a MAOP of 15 MPa, using X120 instead of X80 results reduces the capital costs with 7–8%. As a consequence, the highest available steel grades are selected for onshore pipelines transporting liquid CO₂, namely X80 in the short term, X100 in the medium term and X120 in the long term. Offshore pipelines remain mainly based on lower steel grades because for these pipelines the thickness should be minimal 2.5% of the outer diameter. Also pipelines transporting gaseous CO₂ do not benefit from higher steel grades, due to the low operation pressure.

A cost minimization model was developed which include initial compression, pipeline, steel grades and pumping stations to determine the most cost-effective configuration for CO₂ pipeline transport. Slightly different methodologies were developed for point-to-point pipelines and simple networks over one or multiple kinds of terrain (sparsely populated, populated and offshore). Fig. 15 summarizes the key conclusions for each type of pipeline.

From a chain perspective, gaseous CO₂ transport can have lower levelized costs than liquid CO₂ transport point-to-point pipelines as well as for simple networks. Gaseous CO₂ transport seems to be especially interesting for small CO₂ mass flows on a close distance from a depleted gas field with a low reservoir pressure. The reason for this is that the lower initial compression costs compensate the higher pipeline costs. However, for storage reservoirs with a high reservoir pressure (like aquifers), liquid CO₂ transport is more cost-effective than gaseous CO₂ transport.

For offshore pipelines, the optimal inlet pressure is directly influenced by the length, because in this case pumping stations are

	Point-to-point pipelines	Simple network
Single terrain	- Gaseous CO₂ transport can result in lower levelized costs than liquid CO₂ transport, if the CO₂ is stored in reservoirs with a low pressure. Gaseous CO₂ transport is especially interesting for small CO₂ mass flow and short distances. - Liquid onshore CO₂ transport: $P_{inlet} = 9\text{--}13\text{ MPa}$; $\Delta P = 15\text{--}45\text{ Pa/m}$ - Liquid offshore CO₂ transport: $\Delta P = 10\text{--}50\text{ Pa/m}$, - Gaseous transport: $\Delta P = 5\text{--}20\text{ Pa/m}$	- All four network options result for some combination of mass flow and length in the lowest levelized costs. - However, transporting gaseous CO₂ and compress it to a liquid just before the trunkline is only interesting in a few cases and this is mainly due to the assumptions made about design pressure drop. - The maximum distance of feeders without a
Multiple terrain	- The levelized costs increase with 8–18% for a 100 km pipeline when the terrain change from sparsely populated to populated terrain. - For longer pipelines the cost increase is slightly less due to similar costs for pumping stations in different types of terrain. - The ΔP is slightly higher on populated than on sparsely populated terrain.	pumping stations before the offshore trunk-line increases with decreasing trunkline lengths. For instance, the maximum distance is 75 km for a 100 km and 40 km for a 1,000 km long offshore trunkline. - Break-even distances are independent of mass flow, but variation is present due to the limited NPS available. - Gaseous CO₂ transport in whole the network can be cost-effective for short networks and limited mass flows.

Fig. 15. Main conclusions of the cost minimization model, divided in point-to-point pipelines and simple networks over a single or multiple terrain types.

not an option. The inlet pressure can become up to about 25 MPa. For even longer pipelines, a larger diameter is selected to limit the specific pressure drop. For instance, for a pipeline transporting 300 kg/s liquid CO₂ (about 9 Mt/y) the optimal inlet pressure is 12 MPa for 100 km and 22 MPa for 340 km, leading to an average pressure drop of about 45 Pa/m. If the distance increases to 350 km, a larger diameter is selected (namely 0.61 m instead of 0.41 m) to limit the inlet pressure (to 14 MPa) and the specific pressure drop (to 14 Pa/m). This results in levelized costs (excluding compression) of 1.4 and 4.8 €/t for transporting 300 kg/s CO₂ over 100 and 350 km, respectively. For larger mass flow rates, the levelized costs are decreasing.

The assessment also indicates that the optimal diameter, and hence the specific pressure drop, is mainly influenced by a decrease in the labor costs and interest rate or a change in electricity costs. Changes in the steel costs affect primarily the optimal inlet pressure of the pipeline, and consequently the distance between pumping stations. Furthermore, the sensitivity analysis showed that changes in the electricity costs have the largest influence on the total levelized costs due to the high compression costs. However, if the compression costs are excluded from the levelized costs, the labor costs and interest rate have the largest impact followed by the steel and electricity costs.

There are still a number of points in which way the cost minimization model can be improved further.

- A network design will not only be based on economic optimums but will also depend on legislation, operational preference, risk

and safety aspects. Furthermore, reliability considerations will be important.

- The current version of the model was not linked to topographical maps and different types of terrain (offshore, sparsely populated and populated) serve as a manual input.
- Impurities were not taken into account and these would influence the phase envelope and the physical properties of the CO₂. Hence, it will affect the capacity, the minimum pressure and the material requirements of the pipeline.

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Appendix A. Appendix

Additional equations and methodology

In the methodology section not all formulas were mentioned for clarity reasons. The Darcy–Weisbach friction factor (Eq. (18)) and Reynolds number (Eq. (19)) are used for calculating $ID_{calculated}$. The next available OD_{NPS} is selected with respect to $ID_{calculated}$. For

OD_{NPS} , the required thickness is determined and ID_{NPS} is calculated with Eq. (20):

$$1/\sqrt{f} = -1.8 \log \left[(\varepsilon/ID_{NPS}/3.7)^{1.11} + 6.9/Re \right] \quad (18)$$

$$Re = \frac{\rho * ID_{NPS} * v}{\mu} \quad (19)$$

$$ID_{NPS} = OD_{NPS} - 2t \quad (20)$$

where f is the Darcy–Weisbach friction factor; ε is the roughness height ($=50 \times 10^{-6}$ m); ID_{NPS} is the inner diameter of the NPS pipeline (m); Re is Reynolds number; ρ is the density (kg/m^3); v is the actual velocity (m/s); μ is the viscosity³¹ (Pa s); t is the thickness (m) and OD_{NPS} is the outer diameter of the nominal pipe size (m).

When ID_{NPS} is smaller than $ID_{calculated}$, the next OD_{NPS} is selected and the required thickness for this diameter is determined. This process is repeated until ID_{NPS} is larger than $ID_{calculated}$. Subsequently, the capital costs of the pipeline are calculated. The thickness calculation process is conducted again with a higher steel grade and also for this steel grade the costs of the pipeline are determined. When all steel grades are covered, the combination of steel grade, ID_{NPS} and thickness resulting in the lowest capital costs is selected for a given case.

Since ID_{NPS} can be larger than $ID_{calculated}$ the distance between and number of pumping stations may be too pessimistic. Therefore, the actual pressure drop, distance between and number of pumping stations are calculated again for all liquid cases with Eqs. (21), (22) and (23), respectively. As a consequence, the pumping stations are not evenly distributed over the pipeline and the last pumping station is placed relatively close to the sink.³² The capacity of the last pumping station is adapted such that the outlet pressure of the pipeline match the design outlet pressure (see Eq. (24)). If no pumping stations are installed on the pipeline, the initial pressure inlet and compression capacity is adapted in a similar way. This is also done for the gaseous cases with Eq. (25). These adaptations have in principle consequences for the MAOP of the pipeline. This small deviation is, however, not taken into account in this analysis:

$$\Delta P_{act} = \frac{8 * f * m^2}{\pi^2 * \rho * ID_{NPS}^5} \quad (21)$$

$$L_{pump} = \frac{P_{inlet} - P_{outlet}}{\Delta P_{act}} \quad (22)$$

$$N_{pumps} = \left\lceil \frac{L}{L_{pump}} \right\rceil \quad (23)$$

$$P_{outlet_last_pump} = P_{outlet} + (L - L_{pump} * N_{pumps}) * \Delta P_{act} \quad (24)$$

$$P_{outlet_adapted_gas} = \left(\frac{16 * Z_{ave}^2 * R * T_{ave}^2 * m * f * L}{\pi^2 * ID_{NPS}^5 * M * Z_{ave} * T_{ave} * R} + \frac{2 * g * P_{ave}^2 * M * \Delta z}{Z_{ave} * T_{ave} * R} + P_{inlet}^2 \right)^{0.5} \quad (25)$$

where ΔP_{act} is the actual pressure drop (Pa/m); f is the Darcy–Weisbach friction factor; m is the CO_2 mass flow (kg/s); ρ density (kg/m^3); ID_{NPS} inner diameter of the NPS pipeline (m); L_{pump}

maximum distance between pumping stations (m); P_{inlet} pressure inlet (Pa); P_{outlet} pressure outlet (Pa); N_{pumps} number of pumping stations on the pipeline; L length of the pipeline (m); $\lceil \dots \rceil$ is the largest integer not greater than the enclosed ratio; $P_{outlet_last_pump}$ is the pressure outlet of the pumping station closest to the sink (Pa); $P_{outlet_adapted_gas}$ is the actual pressure outlet of the compressor (Pa).

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ijggc.2013.12.016.

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³¹ The viscosity is based on the lowest pressure in the pipeline (P_{outlet}) to take a conservative approach.

³² In reality, the first pumping station would be installed relatively close to the source and the last pumping station would be placed far away from the sink. In this way, the minimum operation pressure is only approached at the outlet. The reason for this is that the density of the CO_2 flow is somewhat higher meaning that the pressure drop is less and the compression requirement is lower. Nevertheless, as in this study the density is assumed to remain constant, the optimal configuration would not change and the current assumption about placement is easier to model.

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